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Universal properties from quantum many-body dynamics

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The Loschmidt echo—the amplitude for a quantum state to return to itself—provides a route to the universal properties of a critical many-body system. Continued from real to imaginary time, it becomes the partition function of a conformal field theory on a strip, and its universal data—the central charge and the operator content—are encoded in the spectrum of a transfer matrix, obtained by contracting the space–time tensor network along the space direction. We apply this framework to a transverse-field Ising model extended by a next-nearest-neighbour coupling: a non-integrable chain that remains critical and in the Ising universality class over a wide range of that coupling. After a quench to the critical point, the temporal entanglement that governs the cost of the transverse contraction grows only logarithmically with the evolution time, so the method remains efficient at times where conventional evolution algorithms are limited by the entanglement barrier. We develop a translation-invariant matrix product operator, accurate to second order in the time step, that represents the time evolution of the model, and use two power methods to compute the leading eigenvectors and eigenvalues of the resulting non-Hermitian transfer matrix. Within the accessible time windows, our results are consistent with the central charge and operator content predicted by the Ising universality class.

Keywords: Loschmidt Echo, Quantum Quench, Tensor Networks, Transverse Contraction, Temporal Entanglement, Conformal Field Theory, Criticality

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Generative AI tools were used during the preparation of this thesis. Anthropic's Claude models assisted with brainstorming, preliminary drafts, debugging numerical code, and automating repetitive tasks, including the production of figures and diagrams. OpenAI's ChatGPT and Google's Gemini assisted with the organisation and clarity of the text and with identifying typographical errors. All AI-assisted material was reviewed, verified, and revised by the author, who takes full responsibility for the content of the thesis.

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Acronyms

ANNNI axial next-nearest-neighbour Ising . . . 2
CFT Conformal Field Theory . . . 1
DMRG Density Matrix Renormalisation Group . . . 1
FSM finite-state machine . . . III
JW Jordan–Wigner . . . 9
MPO Matrix Product Operator . . . 2
MPS Matrix Product State . . . 1
NN nearest-neighbour . . . 11
NNN next-nearest-neighbour . . . 2
RDM Reduced Density Matrix . . . 7
RTM Reduced Transition Matrix . . . III
SVD Singular Value Decomposition . . . 28
TN Tensor Network . . . 1
tMPO temporal Matrix Product Operator . . . 6
tMPS temporal Matrix Product State . . . 13

1 Introduction and Motivation

Understanding the evolution of a closed quantum system driven far from equilibrium is a central problem in modern many-body physics. Although such dynamics are difficult to simulate, they can reveal universal information at a critical point. In particular, the real-time evolution of a simple initial state can encode the central charge and operator content of the underlying conformal field theory [1], even though these are equilibrium properties of the low-energy spectrum, normally extracted from the ground state. This route is especially valuable for non-integrable models, where analytical results are scarce.

Exact numerical simulations are severely limited by the exponential scaling of the Hilbert space, which makes representing a generic many-body wavefunction intractable. To circumvent this bottleneck, the community has turned to Tensor Networks (TNs), a framework that efficiently compresses quantum states by using entanglement as the organising principle. For one-dimensional systems, Matrix Product States (MPSs) combined with algorithms such as the Density Matrix Renormalisation Group (DMRG) have become the state-of-the-art approach for computing ground states [2–4]. Their efficiency comes from the area law of entanglement, which holds for ground states of one-dimensional systems with local gapped Hamiltonians: the entanglement entropy across a bipartition saturates to a constant rather than growing with the subsystem size [5]. Since the required MPS bond dimension is controlled by this entanglement, the ansatz naturally targets the low-entanglement corner of Hilbert space where these ground states lie.

Simulating dynamics, however, runs into the so-called entanglement barrier [6, 7]. After a quantum quench, the entanglement entropy typically grows linearly in time, approaching volume-law behaviour [8]. Although highly optimised algorithms such as time-evolving block decimation (TEBD) [9] and the time-dependent variational principle (TDVP) [10] are effective at short times, capturing this growth requires a bond dimension that increases exponentially with time, making conventional simulations intractable at late times [11, 12].

Transverse contraction [13–15] tackles this problem by adopting a different perspective: rather than evolving a state forward in time as in the Schrödinger picture, it represents the full dynamics as a two-dimensional tensor network and contracts it along the original spatial direction. The computational cost is then controlled by the temporal entanglement entropy [16–19], defined through a Schmidt decomposition across a cut in time rather than in space. Although its physical interpretation remains an open question [20], it is computationally well defined. For a generic quench, temporal entanglement may grow as rapidly as spatial entanglement, but transverse contraction can still be useful for specific dynamical quantities.

The central quantity we compute throughout is the Loschmidt echo for an infinite, translation-invariant system: the amplitude for a state to return to itself after evolving for a time T [21]. It measures how quickly the evolved state departs from, and how coherently it returns towards, its initial condition, providing a natural probe of relaxation after a quench. Because the echo is a single space–time amplitude, transverse contraction computes it efficiently, contracting the network directly with a power method [1].

At a conformally invariant critical point, the echo also provides direct access to universal data such as the central charge of the underlying Conformal Field Theory (CFT). In the transverse picture, it is governed by a spatial transfer matrix whose spectrum is predicted by the CFT and which becomes effectively unitary at long times, a phenomenon known as emergent dual unitarity [1, 22]. Both of these predictions have been verified for the critical Ising and three-state Potts chains [1], along with the corrections that arise because a simulation reaches only finite times [23].

This thesis extends that analysis to the non-integrable regime, studying a self-dual quantum spin chain of the axial next-nearest-neighbour Ising (ANNNI) type in the thermodynamic limit: the transverse-field Ising model with an added coupling of strength p between second neighbours [24, 25]. For any $p > 0$ the chain is longer-ranged and no longer solvable, yet it remains critical and in the Ising universality class over a wide range of p, so the CFT predictions provide a clear target for numerical simulations. We prepare the system in the fully polarised paramagnet state, quench it to the critical point, and use the transverse contraction machinery to compute the Loschmidt echo and the generalised temporal entropies, which we compare with the CFT predictions. This requires two numerical developments. First, we construct a translation-invariant time-evolution Matrix Product Operator (MPO) that is second-order accurate and treats the next-nearest-neighbour (NNN) interactions without requiring prohibitively small time steps. Second, we develop a block power method to resolve the subleading transfer-matrix eigenvalues, from which the boundary operator content is obtained.

The thesis is organised as follows. Section 2 introduces the Loschmidt echo and the CFT predictions for the transfer-matrix spectrum and temporal entropies. Section 3 presents the transverse-contraction framework, Section 4 the ANNNI-type model and its equilibrium properties, Section 5 the time-evolution MPO, and Section 6 the two iterations used to extract the transfer-matrix spectrum. Section 7 compares the numerical results with the CFT predictions. The appendices provide supporting derivations, implementation details, numerical controls, and additional measurements.

2 Universal Data from the Dynamics

At a continuous critical point, a lattice model forgets its microscopic details and is described by a CFT, characterised by a small set of universal numbers: the central charge c and the scaling dimensions of its operators. Every model in the same universality class reproduces them, which is what makes them good targets for a numerical method. This section relates the Loschmidt echo to a boundary CFT, derives the four predictions that Section 7 tests, and separates them from the non-universal lattice quantities, which we remove or measure independently.

2.1 From the Loschmidt echo to a boundary CFT

The object we compute is the Loschmidt echo,

$$\mathcal{L}(T) = \langle \Psi_0 | \, e^{-iHT} \, | \Psi_0 \rangle \, . \tag{1}$$

For an infinite, translation-invariant chain prepared in $|\Psi_0\rangle$, this is the return amplitude after evolving for a time T with the critical Hamiltonian. We take $|\Psi_0\rangle$ to be a product state, whose choice fixes the boundary condition discussed below.

To connect this amplitude with the continuum theory, we rotate the evolution time to the imaginary axis, $T \to -i\tau$ (a Wick rotation [26]). The real-time propagator e^{-iHT} then becomes the Euclidean weight $e^{-\tau H}$, the same operator that generates the thermal partition function of a two-dimensional statistical system. The matrix element in Eq. (1) can consequently be written as a Euclidean path integral on a strip of width τ, with both edges fixed by $|\Psi_0\rangle$. A state that can terminate a path integral in this way, without introducing any scale of its own, is what a CFT calls a conformal boundary state [27]. The echo is therefore the partition function of the critical theory on a strip whose width is the (imaginary) evolution time and whose edges carry a conformal boundary condition.

The rotation is used only to obtain the conformal predictions, which are derived on the Euclidean strip; they are continued back to real time before being compared with the simulation. Figure 1 shows this construction, first as a tensor network and then as the strip it represents.

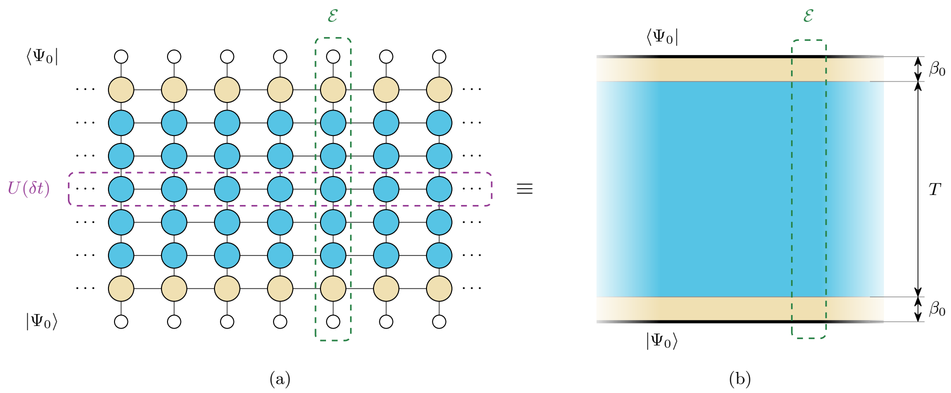

Figure 1: **The Loschmidt echo as a tensor network and as a conformal strip.** (a) Trotterising the evolution with a step δt expresses the echo as the contraction of a two-dimensional network. Open circles are the product state closing the network at the two ends of the time direction; each row of filled circles is one Trotter layer $U(\delta t)$, and the cream rows at top and bottom represent the imaginary-time cooling β_0 that brings the bare product state towards a conformal boundary condition. (b) The same object after the Wick rotation, in the continuum. The heavy lines are the edges where $|\Psi_0\rangle$ closes the path integral, the cream bands are the cooling intervals and the blue bulk is the real-time evolution, so the strip spans $T + 2\beta_0$ in the time direction and is infinite in the spatial one. In both panels the dashed column marks the transfer matrix $\mathcal{E}$, whose construction is developed in Section 3.

Two features of this construction are important for the numerics. First, a bare product state carries weight on high-energy configurations and is not itself a conformal boundary state. We therefore cool it in imaginary time for a short interval β_0, which suppresses that content and brings the state into the regime where the CFT applies. On the lattice, the cooling contributes $N_\beta = 2\beta_0/\delta t$ imaginary-time sites split between the two ends, so the strip spans $T + 2\beta_0$ in the time direction. Second, the initial state is the boundary condition of the strip: each $|\Psi_0\rangle$ realises a different conformal boundary condition, and that choice selects which operators appear in the transfer-matrix spectrum.

In the Euclidean description, space and imaginary time are coordinates of the same two-dimensional geometry, so either can be chosen as the direction of propagation. In the transverse picture, we exchange their roles and contract the network along the original spatial direction. This defines a transfer matrix $\mathcal{E}$ that propagates from one spatial column to the next while acting on states supported on a temporal slice. Its tensor-network construction is developed in Section 3.

In two dimensions, conformal symmetry constrains both the spectrum of $\mathcal{E}$ and the entanglement across a cut in the strip [28, 29]. These two observables provide independent routes to the universal data; Appendix A summarises the conformal mappings used to derive them.

2.2 The universal predictions

Confining a CFT to an infinite strip of transverse size β shifts its ground-state energy by an amount set by the central charge [30]. The excited levels above this shift are organised by the operator scaling dimensions x_i [31]. The transfer matrix inherits this structure [1],

$$\mathcal{E} = e^{-a\hat{H}} = \exp\Big[-\big(-\kappa + \pi L_0\big)\, g\Big], \qquad \kappa = \frac{\pi c}{24}, \tag{2}$$

where L_0 is the operator whose eigenvalues list the scaling dimensions, κ is the Casimir number, and $g = a/\beta$ is the ratio of the column spacing to the strip width, with $a = 1$ in lattice units. The width is complex, $\beta = v(2\beta_0 + iT)$, because the regulator contributes imaginary time while the quench contributes real time. Substituting this width and expanding for $\beta_0 \ll T$ shows that the universal contributions enter the eigenvalue phases at order $1/T$ but affect their moduli only at order β_0/T^2 (Appendix A.2). A single diagonalisation can therefore determine both c and the dimensions x_i from the phases.

The second route uses entanglement. In a CFT, the Rényi entropies follow from correlators of twist operators whose scaling dimension is proportional to c [32]. In the transverse setting, the cut separates earlier from later times rather than dividing space, and its two sides do not belong to a single quantum state. The entropies are instead built from the left and right temporal MPSs of Section 3; they are therefore generalised entropies [20] and are complex in general. For a cut at time t [1, 23],

$$S_n^{\text{gen}}(t) = s_0 + \frac{i\pi c}{24}\Big(1 + \frac{1}{n}\Big) + \frac{c}{12}\Big(1 + \frac{1}{n}\Big)\, W(t, T), \tag{3}$$

with the chord variable

$$W(x, \ell) \;=\; \log\left[\frac{2\ell}{\pi}\, \sin\frac{\pi x}{\ell}\right], \tag{4}$$

where x is the position of the cut and ℓ is the extent of the system. Evaluating W on the interval $(0, T)$ makes the evolution time play the role of the subsystem size in the spatial formula. Because the quench is critical, the entropy grows only as $\log T$, and the bond dimension required for the contraction grows only polynomially with T [11, 12]. We use the Rényi-2 member throughout because it requires only two copies of the system, which also makes it experimentally accessible [33]. The equilibrium calculations of Section 4 instead use the ground-state von Neumann entropy, which DMRG provides directly from the entanglement spectrum.

Equations (2) and (3) now yield four predictions.

Emergent dual unitarity. Equation (2) fixes how the spectrum evolves with T, and predicts that the moduli of the leading eigenvalues approach a common value while their phases stay distinct (see Appendix A.2 for details),

$$\frac{|\mu_i(T)|}{|\mu_0(T)|} = 1 - \frac{b_i}{T^2} - \cdots \xrightarrow[T\to\infty]{} 1, \qquad b_i = \frac{2\pi\beta_0\,(x_i - x_0)}{v}. \tag{5}$$

This indicates that the temporal transfer matrix becomes unitary up to an overall non-universal normalisation [1, 22]. We show this behaviour through the trajectory of $\mu_0(T)$ in the complex plane, which should trace a circle whose coupling-dependent radius reflects the non-universal normalisation.

The central charge from the leading eigenvalue. We work with the logarithms of the transfer-matrix eigenvalues, $\lambda_i = \log(-\mu_i)$, so that $\operatorname{Im}\lambda_i = \arg(-\mu_i)$ is the phase of μ_i measured from the negative real axis. The minus sign shifts every phase by the same constant π, which cancels in phase differences and is absorbed by a constant in the fits. The central-charge and boundary-gap relations are expansions in $1/T$ of these logarithms, rather than of the eigenvalues themselves. Substituting the strip width into Eq. (2) and expanding gives the following relation [1], derived in Appendix A.2:

$$\frac{\operatorname{Im}\lambda_0}{T} = A + \frac{B}{T} + \frac{C}{T^2} + \frac{D}{T^4} + \cdots, \qquad C = -\frac{\kappa}{v} = -\frac{\pi c}{24\, v}. \tag{6}$$

Here A is a non-universal coefficient, D absorbs the subleading finite-time corrections, and $B = -\pi$ is the branch constant introduced by $\lambda_0 = \log(-\mu_0)$, while the coefficient of $1/T^2$ carries the Casimir number divided by the velocity. Once v has been measured independently (Section 4.2), the relation $c = 24\,v\,|C|/\pi$ determines the central charge from the eigenvalue phase alone. The phase is dominated by the non-universal term A, which grows linearly with the evolution time, while the universal contribution is suppressed relative to it by two powers of $1/T$. Isolating C therefore requires phases measured over a wide range of evolution times. In practice we fit the unwrapped phase with B pinned to $-\pi$, leaving A and C free.

The boundary dimensions from the phase gaps. Each subleading eigenvalue sits at a phase distance from the leading one that measures a boundary dimension,

$$\mathrm{Im}(\lambda_i - \lambda_0) = \frac{\pi\,(x_i - x_0)}{v\,T}. \tag{7}$$

Once the sound velocity is known, the gaps determine the dimensions; the first of them gives the lightest, x_1. The spectrum consists of the dominant eigenvalue followed by one subleading eigenvalue for each boundary operator, ordered by increasing phase distance. We refer to it as the conformal tower, and to its members as tower states.

Not every dimension appears. Which ones do is fixed by the pair of boundary conditions at the two ends of the strip [34], and for the Ising class there are three conditions: the order parameter may be free to fluctuate or held fixed in either direction. Each pair admits its own set of operators, and each operator brings its own descendants, spaced by integers [27, 35]. The quench of this thesis uses the same free state at both ends, for which $x_0 = 0$ and the spectrum is $\{0, \frac{1}{2}, \frac{3}{2}, 2, \dots\}$. The lightest dimension also controls how quickly the finite-time corrections decay, so the free boundary at $x_1 = \frac{1}{2}$ converges faster than the fixed one at $x_1 = 2$. Appendix G.2 sets out the selection rule for all four pairs and measures each of them.

The generalised entropy profile. Separating Eq. (3) into real and imaginary parts gives

$$\mathrm{Re}\, S_n^{\mathrm{gen}}(t) = s_0 + \frac{c}{12}\Big(1 + \frac{1}{n}\Big)\, W(t, T), \qquad \mathrm{Im}\, S_n^{\mathrm{gen}} = \frac{\pi c}{24}\Big(1 + \frac{1}{n}\Big). \tag{8}$$

The real part follows a symmetric chord profile: plotted against $W(t,T)$ it becomes a straight line whose slope is $\frac{c}{12}(1 + 1/n)$, derived in Appendix A.1. The imaginary part is independent of t, so it delivers c from a single plateau value with no profile to fit. Both parts contain c, but only the plateau gives a usable estimator on the times we can reach (Appendix G.1). Its finite-time correction decays slowly, as $T^{-1/2}$ for the Rényi-2 entropy used here [23], so the plateau must be extrapolated to long times.

These predictions apply to the continuum theory, whereas a simulation contains additional lattice scales and numerical parameters. The sound velocity v, the entropy offset s_0 and the overall normalisation of the transfer matrix are fixed by the lattice construction rather than by the universality class. We measure v independently in Section 4.2, absorb s_0 into the entropy fits, and remove the normalisation by working with phase differences, so that only universal quantities are compared in Section 7. Section 3 constructs the transfer matrix and the reduced transition matrix used to make these measurements.

3 Time Evolution via Transverse Contraction

Section 2 showed that the Loschmidt echo of a critical quench encodes conformal data in the spectrum of a transfer matrix acting along the spatial direction of the corresponding strip. We now construct this transfer matrix and explain how it is used numerically.

Transverse contraction was introduced for infinite chains by representing the dynamics with MPS extending along the time direction [13, 14], and was subsequently developed to study temporal entropies [1, 18]. Our calculations build on the `ITransverse.jl` implementation [36].

Reading the network along the spatial direction maps the contraction to an eigenproblem for the spatial transfer matrix, whose dominant eigenvectors define the temporal boundary states. These states determine the RTM used to compute the generalised temporal entropies. We finally compare two truncation schemes for controlling the bond dimension of the temporal states.

3.1 The rotated network

The time evolution of a D-dimensional system is represented by a $(D+1)$-dimensional tensor network, with the additional dimension corresponding to time. For the one-dimensional chain studied here, the result is a two-dimensional network with no open physical legs. The echo is a scalar amplitude, so this network is analogous to the partition function of a two-dimensional classical model [37]. Its exact contraction is #P-hard in general [38], so we must use an approximate representation with a controlled bond dimension (Section 3.4).

The tensor-network representation of the echo follows directly from Eq. (1) and is pictured in Figure 1a. The initial and final states are represented as MPSs, while the evolution operator is decomposed into a sequence of MPO layers, one for each Trotter step. Because we require the Hamiltonian to be translation invariant and time independent, the same bulk tensor W appears at every spatial site and time step. Section 5 constructs this tensor for a model with NNN interactions.

In conventional evolution algorithms, successive MPO layers of $U(\delta t)$ are applied to $|\Psi_0\rangle$. The spatial entanglement then grows linearly in time [8], so the bond dimension needed to represent the state grows exponentially [11, 12]. Transverse contraction instead contracts the same network column by column along the spatial direction. The tensors remain unchanged, but their indices exchange roles: the virtual spatial bonds become physical temporal indices, while the physical spatial indices become virtual temporal bonds. Figure 2 illustrates this exchange. The bond dimension of $U(\delta t)$ therefore sets the physical dimension of the temporal problem. Each bulk column defines a temporal Matrix Product Operator (tMPO) $\mathcal{E}$ that acts as a transfer matrix along the original spatial direction, and translation invariance ensures that the same $\mathcal{E}$ is applied at every site.

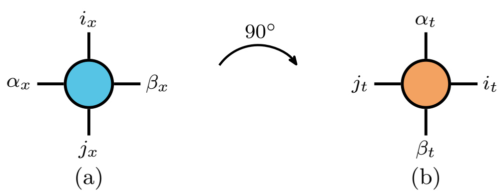

Figure 2: **The rotation exchanges physical and virtual indices.** The bulk tensor W read (a) along space, with (i_x, j_x) physical and (α_x, β_x) virtual, and (b) along time, where (α_t, β_t) are physical and (i_t, j_t) virtual.

These objects provide the lattice representation of the strip geometry introduced in Section 2. The strip is infinite along the spatial direction, and applying infinitely many columns on either side of a cut projects the temporal boundary states $\langle b_L|$ and $|b_R\rangle$ onto the dominant left and right eigenvectors of $\mathcal{E}$ [16, 39], which we write $\langle L|$ and $|R\rangle$. These fixed points are independent of how the chain is terminated far from the cut. They encode the temporal boundary condition imposed by $|\Psi_0\rangle$ at the top and bottom edges. The strip width is set by the number of temporal sites, including the real-time evolution and the cooling sites associated with the regulator β_0. The CFT constrains the spectrum of the column transfer matrix $\mathcal{E}$, while the fixed points $\langle L|$ and $|R\rangle$ are the states from which the echo and the temporal entropies are computed.

The row (Trotter-layer) and column transfer operators have different spectral properties. Each row represents the unitary Trotter step $U(\delta t) = e^{-iH\delta t}$ and therefore has eigenvalues of unit modulus, up to the error of its MPO representation. By contrast, the column transfer matrix $\mathcal{E}$ is non-Hermitian and generally has complex eigenvalues with different moduli. The implications of its non-Hermiticity for the left and right eigenvectors are discussed in Section 5 after the operator has been constructed explicitly.

3.2 The dynamics as an eigenvalue problem

Contracting the network at finite spatial size turns the echo into a sum over the spectrum of $\mathcal{E}$. For a chain of N spatial sites,

$$\mathcal{L}(T) \;=\; \langle b_L|\, \mathcal{E}(T)^N \,|b_R\rangle \;=\; \sum_i c_i\, \mu_i(T)^N \;=\; c_0\, \mu_0(T)^N \Big[1 + \sum_{i\geq 1} \frac{c_i}{c_0}\Big(\frac{\mu_i}{\mu_0}\Big)^N\Big], \tag{9}$$

where μ_i are the eigenvalues of $\mathcal{E}$, assumed diagonalisable, with left and right eigenvectors $\langle \ell_i|$ and $|r_i\rangle$ normalised so that $\langle \ell_i|r_j\rangle = \delta_{ij}$, and $c_i = \langle b_L|r_i\rangle\,\langle \ell_i|b_R\rangle$. The dominant pair is $\langle \ell_0| = \langle L|$ and $|r_0\rangle = |R\rangle$. If μ_0 has the largest modulus, every other contribution is suppressed relative to it by $(|\mu_i|/|\mu_0|)^N$. The thermodynamic limit is therefore determined by the leading eigenvalue, and the intensive Loschmidt rate becomes

$$\ell(T) = -\lim_{N\to\infty} \frac{1}{N} \log|\mathcal{L}(T)| = -\log|\mu_0(T)|, \tag{10}$$

up to $\mathcal{O}(1/N)$ boundary corrections for a finite open chain. The calculation thus reduces to a spectral problem whose cost is independent of the spatial length.

Equation (9) reduces the contraction to the spectrum of $\mathcal{E}$. The modulus of μ_0 gives the Loschmidt rate through Eq. (10), while Section 2 reads the central charge from its phase and the boundary dimensions from the phase gaps. The generalised temporal entropies instead require the dominant left and right eigenvectors. Our calculations must therefore resolve the leading eigenpairs and spectrum of the non-Hermitian operator $\mathcal{E}$, which we obtain with the procedures described in Section 6.

3.3 Temporal states and the RTM

Each application of $\mathcal{E}$ increases the bond dimension of the temporal boundary states. The computational cost is therefore controlled by their temporal entanglement, defined across a cut in the time direction. Section 2.2 showed that after a critical quench the generalised temporal entropies grow logarithmically with the evolution time. We now define the object from which these entropies are computed.

The dominant left and right fixed points of the non-Hermitian transfer matrix are distinct, and the echo depends on their overlap $\langle L|R\rangle$. A Reduced Density Matrix (RDM) formed from either state alone therefore does not contain the required information. The relevant object is instead the RTM constructed jointly from the two temporal boundaries [1, 18],

$$\mathcal{T}_t = \frac{\mathrm{Tr}_{>t}\,|R\rangle\,\langle L|}{\langle L|R\rangle}, \tag{11}$$

where $\mathrm{Tr}_{>t}$ traces over the $T - t$ temporal sites above the cut and leaves the t sites below it open. Thus, $\mathcal{T}_t$ acts on the first t temporal sites. The denominator normalises the matrix so that $\mathrm{Tr}\,\mathcal{T}_t = 1$. Since $\langle L|$ is not the Hermitian conjugate of $|R\rangle$, $\mathcal{T}_t$ is generally non-Hermitian and may have complex eigenvalues. The consequences of this complex spectrum for truncation are discussed in the next subsection. Figure 3 shows the position of $\mathcal{T}_t$ within the strip geometry.

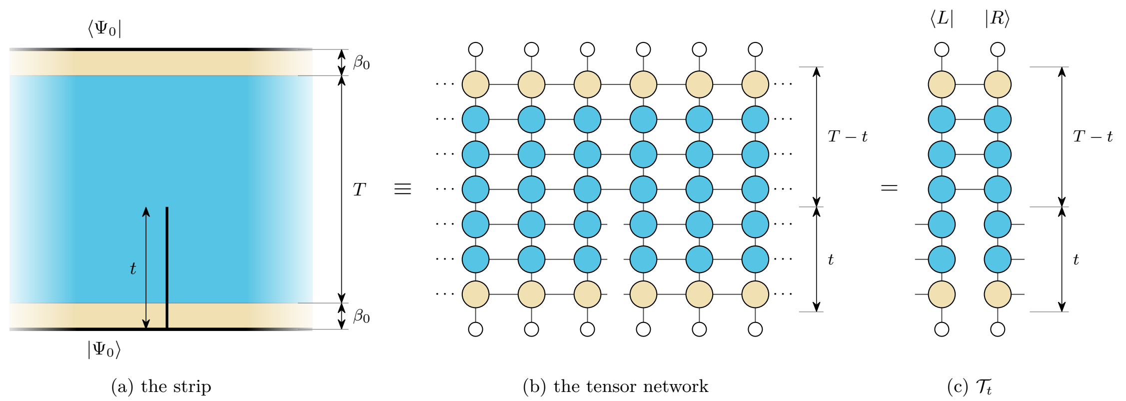

Figure 3: **Position of the reduced transition matrix.** (a) The cut lies on one spatial bond of the strip and extends from the bottom edge to height t; the cream bands at the two ends are the cooling intervals β_0, and the heavy edges carry the initial state $|\Psi_0\rangle$. (b) In the tensor-network representation, the horizontal bonds across the cut remain open over the first t temporal sites, while the two sides remain contracted over the remaining $T - t$ sites. (c) Contracting the columns to the left and right of the cut gives the temporal boundary states. The power methods of Section 6 converge them to the dominant left and right eigenvectors $\langle L|$ and $|R\rangle$ of $\mathcal{E}$. Their contraction over the upper $T - t$ sites gives $\mathcal{T}_t$ in Eq. (11), with the lower t indices left open.

The spectrum of $\mathcal{T}_t$ defines the generalised temporal entropies associated with the transition between the two boundary states. We use the Rényi-2 member of the family introduced in Section 2,

$$S_2^{\mathrm{gen}}(t) = -\log \mathrm{Tr}\left(\mathcal{T}_t^2\right). \tag{12}$$

When $\langle L|$ and $|R\rangle$ differ, the local reduction of $\mathcal{T}_t$ preserves its singular values but not its eigenvalues, so its spectrum cannot be read from a local decomposition (Appendix B). This is the member we can compute: $\mathrm{Tr}(\mathcal{T}_t^2)$ is evaluated by contracting two copies of the RTM, without diagonalising it. The finite-time corrections to S_2^{gen} must be accounted for when comparing with the conformal predictions (Section 2.2).

Once $\langle L|$ and $|R\rangle$ have converged for a target evolution time T, we evaluate Eq. (12) at every internal bond of the temporal chain, corresponding to cuts $t \in (0, T)$. The entropy is generally complex because $\mathcal{T}_t$ is non-Hermitian, so its real and imaginary parts are analysed separately as functions of the normalised cut position t/T.

The region represented by $\mathcal{T}_t$ touches the lower edge of the strip, so only one of its endpoints lies in the interior. This halves the logarithmic coefficient in Eq. (8) relative to an interval with both endpoints in the bulk (Appendix A.1). The entropy-route central charges reported in Section 7 are extracted using this geometry.

3.4 Truncation based on RDM and RTM

We use two complementary truncation schemes based on the RDM and the RTM. The conventional RDM truncation, available in most tensor-network libraries, brings a boundary MPS into mixed canonical form and discards the smallest singular values of its orthogonality centre, giving the optimal low-rank approximation to that state [3, 40]. The gauge-invariant scheme implemented in `ITransverse.jl` [36] instead truncates the RTM of Eq. (11), which depends on both temporal boundaries.

The difference between both schemes is in what they consider as relevant. The RDM keeps the directions in which $|R\rangle$ or $\langle L|$ has the largest weight, regardless of their contribution to $\langle L|R\rangle$. The RTM instead weights each direction by its contribution to this overlap, so a large component of $|R\rangle$ may have little importance if it is orthogonal to the corresponding component of $\langle L|$ and can be discarded [18, 39]. The RTM therefore reaches a given accuracy at a smaller bond dimension, with the RDM truncation being approximately an order of magnitude more expensive in our calculations (Appendix F.5).

This reduction in cost comes at the expense of not having a variational guarantee for RTM truncation. For a positive Hermitian RDM, the eigenvalues are the squared Schmidt coefficients and coincide with its singular values, so retaining the largest gives the optimal approximation at fixed bond dimension. The RTM on the other hand is non-Hermitian: its eigenvalues may be complex or negative and no longer correspond one to one with its singular values, so the truncation criterion carries no variational bound. We therefore use the RTM scheme for production calculations and retain the RDM scheme as a complementary check; Appendix F.5 verifies that the two agree at every evolution time where both were run. Appendix B presents both truncations in detail, including their behaviour under gauge transformations and the conditions for optimality.

4 The ANNNI-type Model

We study a self-dual quantum spin chain that extends the transverse-field Ising model with NNN interactions [25]. For spin-1/2 degrees of freedom, the Hamiltonian is

$$H = -\sum_i \Big[\sigma_i^z\sigma_{i+1}^z + \lambda\,\sigma_i^x \;+\; p\,(\sigma_i^z\sigma_{i+2}^z + \lambda\,\sigma_i^x\sigma_{i+1}^x)\Big], \tag{13}$$

where $\sigma^{x,z}$ are Pauli matrices, λ is the transverse field, and p controls the strength of the additional interactions. The $\sigma^x\sigma^x$ term preserves self-duality under the Kramers–Wannier transformation [25], which fixes the critical point at $\lambda = 1$. We work at this point throughout, so that the conformal predictions of Section 2 apply.

At $p = 0$, Eq. (13) reduces to the integrable transverse-field Ising chain. For $p > 0$, a Jordan–Wigner (JW) transformation no longer maps the Hamiltonian to free fermions, and the model is therefore non-integrable. The origin of the fermionic interactions is detailed in Appendix C.

4.1 Equilibrium criticality and the central charge

We restrict our equilibrium calculations to $0 \le p \le 1.5$, the range over which the model is known to remain critical and in the Ising universality class [25]. We verify the equilibrium universality class over this range using the ground-state entanglement. For a critical open chain of length L, CFT predicts that the von Neumann entropy of a block of length l scales as [32, 40, 41]

$$S_{\mathrm{vN}}(l, L) = \frac{c}{6}\,W(l, L) + k, \tag{14}$$

where W is the chord variable of Eq. (4) and k is a non-universal constant. The von Neumann entropy is well suited to this calculation because DMRG provides the full ground-state Schmidt spectrum directly, and also has weaker finite-size corrections than the higher-order Rényi entropies [42, 43].

For each value of p, we obtain the ground state of a chain with $N = 400$ sites and evaluate the entropy across every bond. Equation (14) predicts a linear dependence on $W(l, L)$, so the entanglement profile of a single ground state provides an estimate of the central charge. We fit only the middle part of the chain, using the cuts satisfying $0.25L < l < 0.75L$, where boundary corrections are smallest. The resulting profiles are shown in Figure 4.

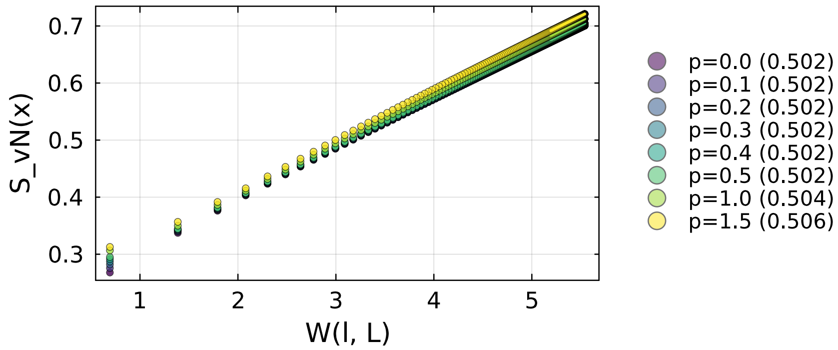

Figure 4: **Effect of NNN interactions on the equilibrium central charge.** Ground-state entanglement profiles at $\lambda = 1$ and $N = 400$, for the eight couplings $p = 0$ to 1.5, plotted against the chord variable W of Eq. (4), which is smallest for cuts near the chain ends. Circles are the entropies at each cut; the solid line through each profile is the fit over the middle half of the chain, whose central charge is given in the legend. The linear dependence on W gives a value close to the Ising $c = 1/2$ at every p, with a small common offset from finite-size corrections.

The profiles follow the conformal scaling law at every coupling, and the fitted central charge barely changes with p. Its small common offset from the exact value $c = 1/2$ is consistent with residual finite-size corrections to Eq. (14), since a similar offset is already present at the exactly solvable point $p = 0$. The additional interactions therefore do not change the universality class over the range considered. This agrees with previous finite-size scaling studies of the same chain [25], and with an independent determination that extracts the conformal data directly from the lattice Hamiltonian [44].

The profiles separate at small W, where the cuts lie closest to the open boundaries. Equation (14) is less accurate in this region because the boundaries introduce corrections that depend on the microscopic interactions and therefore on p, which is why these cuts are left out of the fit. Towards the centre of the chain, these boundary effects weaken and the profiles converge towards the common linear behaviour predicted by CFT. Appendix D.1 gives an independent estimate from the finite-size scaling of the mid-chain entropy and a cross-check using the Rényi-2 entropy.

4.2 Sound velocity

The NNN interaction changes how fast excitations travel along the chain. At an Ising-class critical point, the low-energy dispersion is linear, $\omega(k) = v|k|$, so the sound velocity v fixes the relativistic scale of the continuum theory [29]. It enters three of the four predictions of Section 2 explicitly, because those predictions are written in terms of lengths on the conformal strip and v is what turns an evolution time into such a length [30, 31]; in the entropy profile it only shifts the non-universal offset. The velocity must therefore be measured independently before the predictions can be compared with the dynamical data. At $p = 0$ the chain is a free-fermion model with the exact dispersion $\omega(k) = 4|\sin(k/2)|$, hence $v = 2$ [45]; for $p > 0$ we determine v numerically.

We extract $v(p)$ from the equilibrium dispersion of the same chain, Eq. (13) at $\lambda = 1$, placed on rings of N sites with periodic boundary conditions and diagonalised exactly. The momenta on a ring are quantised, so the slope follows from the energy of the lowest excitation carrying the smallest non-zero momentum [25]. Because exact diagonalisation is restricted to small rings, we repeat this at five lengths and extrapolate with a $1/N^2$ dependence, a form fixed empirically at the integrable point where the answer is known (Appendix D.2). This procedure reproduces the exact result at $p = 0$ and is then applied at each non-zero coupling p.

The velocity grows steadily from 2.00 at $p = 0$ to 10.80 at $p = 1.5$, as shown in Figure 5. Although the velocity is non-universal, it is fixed by the model; we determine it from the equilibrium spectrum and use the independently extrapolated value at each coupling in all subsequent conversions. Appendix D.2 describes the dispersion calculation, extrapolation and validation in detail.

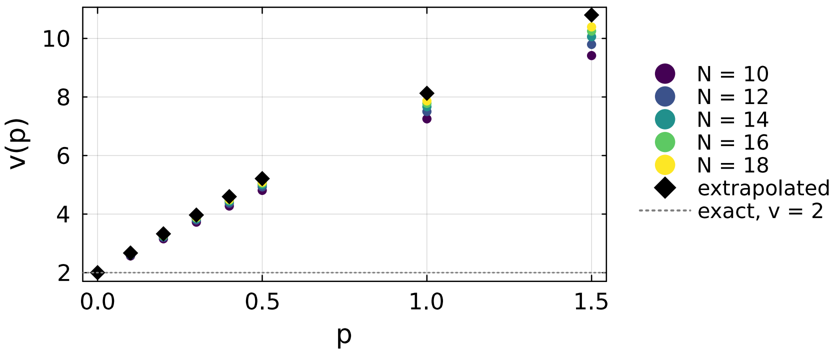

Figure 5: **The sound velocity grows with the NNN coupling.** $v(p)$ at $\lambda = 1$, from the equilibrium dispersion of periodic rings of $N = 10$ to 18 sites (coloured circles), together with the values extrapolated to the thermodynamic limit (black diamonds), which are the ones used throughout. The dotted line marks the exact Ising value $v = 2$ at $p = 0$.

5 Building the Time-Evolution MPO

For nearest-neighbour (NN) Hamiltonians, the evolution operator $U(\delta t) = e^{-iH\delta t}$ is usually represented as a Trotter–Suzuki circuit of two-site gates, giving rise to the usual brick-wall structure [12]. However, an interaction between sites i and $i + 2$ cannot be assigned to a single bond within this pattern. To preserve the one-site translation invariance required by the transverse contraction, as discussed in Section 3.1, we instead express the Hamiltonian as a finite-state machine (FSM) and exponentiate the resulting representation.

5.1 The Hamiltonian as a FSM

In the FSM representation [46, 47], the Hamiltonian is encoded as an MPO whose local tensor has a block-upper-triangular form [48]. Two identity blocks propagate the initial and final states, while the on-site block D generates local terms. The block row C initiates interactions, the block column B terminates them, and the memory block A

propagates unfinished interactions between their endpoints. The virtual index therefore records whether an interaction has not yet begun, is being propagated, or has terminated. For more complicated Hamiltonians, this block structure can be generated automatically from the operator [49]. For the ANNNI-type model of Eq. (13), the local tensor has bond dimension $D_W = 5$ and takes the form

$$W_H = \begin{pmatrix} \mathbb{1} & C & D \\ 0 & A & B \\ 0 & 0 & \mathbb{1} \end{pmatrix} = \begin{pmatrix} \mathbb{1} & -\sigma^z & -p\lambda\,\sigma^x & -p\,\sigma^z & -\lambda\,\sigma^x \\ 0 & 0 & 0 & 0 & \sigma^z \\ 0 & 0 & 0 & 0 & \sigma^x \\ 0 & \mathbb{1} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \mathbb{1} \end{pmatrix}. \tag{15}$$

Figure 6 represents Eq. (15) through the allowed paths between the five virtual states. Each node denotes a value of the virtual index, while an arrow from state a to state b, labelled by an operator O, represents the tensor entry $(W_H)_{ab} = O$. The FSM moves from left to right along the chain, beginning in state 1 and ending in state 5.

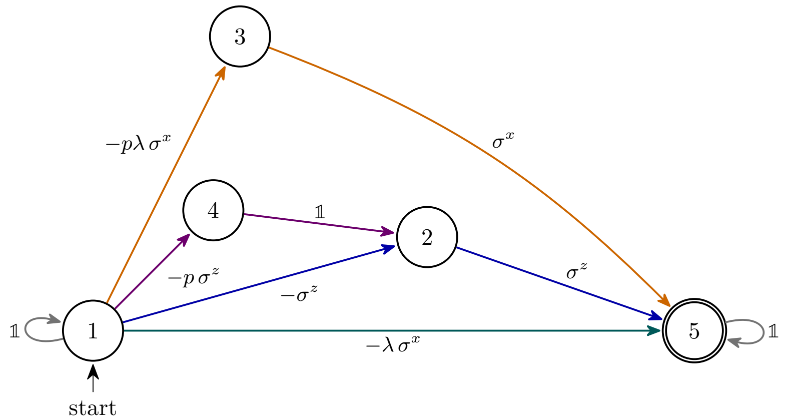

Figure 6: **The Hamiltonian as a FSM.** The nodes are the five virtual states of Eq. (15), and every path from state 1 to state 5 (double circle) generates one term of the Hamiltonian, applying the operator that labels each arrow. Colours mark which term is generated: the transverse field in teal, the NN couplings in blue ($\sigma^z\sigma^z$) and orange ($\sigma^x\sigma^x$), and the NNN coupling in purple, whose path closes through the same final σ^z arrow as the blue pair. Grey loops apply identities before and after the term.

The loop $1 \to 1$ applies identities until a Hamiltonian term is initiated. A direct transition $1 \to 5$ applies the on-site field and completes the term at the same site. An NN interaction instead follows either $1 \to 2 \to 5$ or $1 \to 3 \to 5$: the first transition applies the operator at site i and records whether a σ^z or σ^x endpoint remains to be applied; the second applies the corresponding operator at site $i+1$ and completes the interaction. The NNN term follows $1 \to 4 \to 2 \to 5$: the first transition applies σ^z at site i, the second inserts an identity at site $i+1$, and the third applies σ^z at site $i+2$. After state 5 is reached, the loop $5 \to 5$ applies identities along the remainder of the chain. Zero entries in W_H correspond to absent arrows, so no other transitions are allowed.

Because the FSM is directional, its local tensor is asymmetric under interchange of the virtual indices. The exponentiated operator inherits this structure, as does the spatial transfer matrix after moving to the transverse picture. The resulting operator $\mathcal{E}$ is generally non-Hermitian and non-normal, $\mathcal{E}\mathcal{E}^\dagger \neq \mathcal{E}^\dagger\mathcal{E}$. Its left and right eigenvectors are therefore not related by Hermitian conjugation, and we compute the two temporal boundary states separately.

5.2 Exponentiating the FSM

We seek an approximation $W(\tau) \approx e^{\tau H}$, with $\tau = -i\delta t$, built from the blocks of W_H. Because our Hamiltonian carries interactions at two different ranges, we use the construction of Van Damme et al. [50], which organises the expansion by the spatial overlap of the local Hamiltonian terms. Contributions with disjoint support factorise and are kept to all orders, so only connected clusters of overlapping terms have to be reproduced explicitly. Appendix E.1 shows how the operator is built from the machine of Figure 6.

We use its second-order member, denoted *VD2*, whose virtual bond dimension is 13 for the model of Eq. (13). As noted in Section 3.1, the virtual bond dimension of the time evolution MPO becomes the local physical dimension of the temporal chain after moving to the transverse picture. Consequently, *VD2* carries a larger local dimension than cheaper constructions like the ones of Zaletel et al. [51], which are only first order in a single application. For a given accuracy, the latter thus demand a smaller time step, while *VD2* can hold the same accuracy at a bigger δt. The larger step shortens the temporal chain at fixed physical time and lowers the number of truncation steps, which is what makes the long-time calculations of Section 7 affordable. Appendix E gives the explicit tensor and a measurement of the convergence order of each operator.

The temporal transfer matrix is assembled from a single tensor of the time evolution MPO, repeated along the chain, so the tensor we extract must be one that a site in the bulk would carry, with every interaction present. We therefore exponentiate the Hamiltonian on a chain long enough for its central site to have all of its partners inside it, five sites for an NNN model, and take the tensor at the centre. Every result in this thesis uses this construction.

6 Extracting the transfer-matrix spectrum

Section 3 reduced the dynamical calculation to the problem of determining the spectrum of the transfer matrix $\mathcal{E}$. We use two iterative methods, chosen according to the spectral information required by each observable and their computational cost.

The right and left eigenvectors of $\mathcal{E}$, written $|r_i\rangle$ and $\langle \ell_i|$ as in Section 3, satisfy

$$\mathcal{E}\,|r_i\rangle = \mu_i\,|r_i\rangle\,, \qquad \langle \ell_i|\,\mathcal{E} = \mu_i\,\langle \ell_i|\,. \tag{16}$$

Since $\mathcal{E}$ is non-normal (Section 5.1), its right and left eigenvectors differ and must be computed separately. They form a biorthogonal set, such that $\langle \ell_i|r_j\rangle \propto \delta_{ij}$. Throughout, the bra notation does not imply complex conjugation: $\langle \ell_i|$ denotes a row vector, and $\langle \ell_i|r_j\rangle$ is the direct contraction of the two temporal networks. We implement the second part of Eq. (16) by propagating the left states with $\mathcal{E}^{\mathsf{T}}$, since as column vectors it reads $\mathcal{E}^{\mathsf{T}}\,|\ell_i\rangle = \mu_i\,|\ell_i\rangle$.

6.1 The single-vector iteration

The first option we use is the power method implemented in `ITransverse.jl` [36]. Starting from a random complex temporal Matrix Product State (tMPS), we repeatedly apply the transfer matrix and normalise after each step. As shown in Figure 7, every application multiplies the temporal bond dimension by the tMPO bond dimension D, so we use either truncation scheme of Section 3.4 to cap the bond dimension to χ, keeping the cost per iteration fixed. We use random initial states because a structured product state might have little overlap with the dominant eigenvector and therefore can converge to a subdominant fixed point.

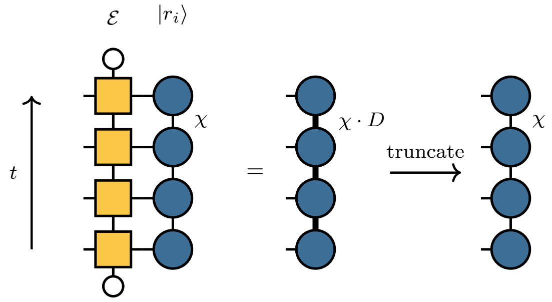

Figure 7: **One application of the transfer matrix to a temporal state.** The tMPS forms a boundary column, while $\mathcal{E}$ is the corresponding column of the space–time network. Applying it raises the temporal bond dimension from χ to χD, where D is the tMPO bond dimension, and the state is then truncated back to χ. The left state is treated in the same way using the transpose.

The iteration converges to the dominant right and left fixed points, from which we obtain the leading eigenvalue as

$$\mu_0 = \frac{\langle L| \mathcal{E} |R\rangle}{\langle L|R\rangle}. \tag{17}$$

Therefore, we use this single-vector method in Section 7 for computing the leading eigenvalue and the associated dominant pair ($\langle L|$ and $|R\rangle$). Obtaining the boundary dimensions x_i, on the other hand, requires access to the subleading eigenvalues of the spectrum to compute the gaps between them.

6.2 The block iteration

To resolve the subleading eigenvalues, we propagate blocks of k right and left states, denoted by $|R_j\rangle$ and $\langle L_j|$. Repeated application of $\mathcal{E}$ drives each block towards the subspace spanned by the k leading eigenvectors, although its individual states do not generally converge to particular eigenvectors and must be separated afterwards. The convergence rate is controlled by the spectral separation between the retained subspace and the discarded eigenvalues, rather than by the gaps within the retained spectrum. The block can therefore converge even as the leading eigenvalues approach the common, constant modulus structure described in Section 2.2.

The individual eigenvalues are then separated by projecting the eigenproblem onto the converged subspace. We approximate a right eigenvector as a linear combination of the current block states,

$$|\psi\rangle = \sum_{j=1}^{k} v_j |R_j\rangle , \tag{18}$$

with coefficients v_j to be determined. Inserting this ansatz into the eigenproblem $\mathcal{E}|\psi\rangle = \theta|\psi\rangle$, with θ the corresponding approximate eigenvalue, produces a residual which we require to vanish when contracted with every member of the left block, $\langle L_i| (\mathcal{E} - \theta) |\psi\rangle = 0$. This two-sided projection is appropriate for a non-normal operator [52] and reduces the full eigenproblem to two $k \times k$ matrices,

$$S_{ij} = \langle L_i|R_j\rangle , \qquad M_{ij} = \langle L_i| \mathcal{E} |R_j\rangle , \tag{19}$$

and to the generalised eigenvalue problem

$$\sum_j M_{ij}\, v_j = \theta \sum_j S_{ij}\, v_j. \tag{20}$$

The overlap matrix S describes the relative orientation of the left and right subspaces, while M is the transfer matrix projected between them. Solving this reduced problem yields approximations θ_j to $\mu_0, \ldots, \mu_{k-1}$ and hence the phase gaps that encode the boundary operator content.

The block iteration can also reconstruct the individual eigenvectors, but this step is computationally expensive. Each eigenvector is formed as a linear combination of the propagated block states, which increases the temporal bond dimension and requires a further truncation to χ. This repeated enlargement and truncation dominates the cost of the eigenvector mode. The single-vector method avoids this operation and provides the dominant eigenvectors at a lower cost, so we use it for determining the dominant pair and the block method only for the subleading eigenvalues. Appendix F gives the algorithm in full, together with its validation against exact diagonalisation.

6.3 Selecting and tracking the eigenvalue and the dominant pair

The block iteration returns k unlabelled Ritz values. At each time, we must identify the leading one, μ_0, and assign the remaining values to the conformal tower. Selecting the value of largest modulus is unreliable because, for a non-normal operator, an unconverged subspace can produce spurious Ritz values outside the spectrum, and their number may increase with the block size [52]. We instead identify the physical branch through continuity in time. We retain candidates whose modulus lies within a few per cent of the previously accepted μ_0 and select the nearest of these in the complex plane. If no candidate satisfies this criterion, we discard the time point. The remaining values are ordered by increasing phase distance from μ_0, as predicted for the conformal tower. We compute each sequence at increasing evolution times and initialise every block with the converged result from the preceding time.

The single-vector iteration returns one eigenvalue per run, so its reliability is assessed across independent seeds rather than among several Ritz values. At short times, all seeds converge to the same left and right fixed points. At longer times, their overlap $\langle L|R\rangle$ can decrease, making the pair ill-conditioned and allowing different runs to converge to different fixed points. We accept an entropy time point only when two independent runs agree and the resulting profile varies smoothly between neighbouring times. The same criterion extends the spectral analysis to longer times. When the independent seeds do not agree, we keep the one whose phase continues the advance established over the preceding times. The new time then enters the sequence only if including its point does not increase the root-mean-square residual of the fit to Eq. (6), so a time is retained only when it is consistent with the trend already established by the earlier ones. Appendix G.1 records these fitting conventions, and Appendix F the numerical controls behind both routes, together with the measured reliability of the two iterations.

7 Results

We test the CFT predictions of Section 2.2 using simulations of the critical ANNNI-type chain, Eq. (13) with $\lambda = 1$, quenched from the polarised state $|X+\rangle$. Extracting the universal data requires a sufficiently long time window, while the computational cost increases rapidly with p. For simplicity, we thus restrict the dynamical calculations to the range from the integrable point, $p = 0$, to $p = 0.5$.

We follow the two routes introduced in Section 6: the entropy route, which requires the dominant eigenvectors, and the spectral route, which uses the leading eigenvalues. Both analyses include the integrable Ising point and the non-integrable cases, allowing us to determine whether the conformal signatures persist after introducing the NNN interaction. Table 1 summarises the resulting universal data.

Unless stated otherwise, the simulations use the second-order *VD2* MPO of Section 5, with time step $\delta t = 0.1$, cooling regulator $\beta_0 = 0.2$ ($N_\beta = 4$), and maximum temporal bond dimension $\chi = 64$; the bond dimensions the two iterations actually reach, together with the remaining numerical controls, are reported in Appendix F.3. We convert the spectral measurements into universal quantities using the independently determined sound velocities $v_\infty(p)$ from Section 4.2. Appendix G.1 gives the fitting conventions and the origin of the uncertainties.

		entropy route		spectral route		
p	v_∞	window	c	window	c	x_1
0	2.00	$4 \le T \le 22$	0.49 ± 0.01	$2 \le T \le 22$	0.45	0.498
0.1	2.67	$4 \le T \le 13$	0.50 ± 0.04	$2 \le T \le 17$	0.47	0.497
0.3	3.97	$2 \le T \le 9$	0.53 ± 0.04	$2 \le T \le 10$	0.49	0.493
0.5	5.21	$2 \le T \le 5$	0.58 ± 0.04	$2 \le T \le 6$	0.53	0.487

Table 1: **Universal data as a function of the NNN coupling.** The entropy-route central charge comes from the extrapolated imaginary plateau of Section 7.1, measured on seed ensembles of independent single-vector runs; its uncertainty is the spread of that extrapolation under resampling of those runs. The spectral central charge comes from the phase curvature of μ_0 through Eq. (6) and x_1 from the first phase gap through Eq. (7), which carry no uncertainty because independent seeds return the same eigenvalues. Each route has its own fitting window, listed beside its result.

7.1 The entropy route

At the integrable point, the generalised temporal entropy of Eq. (12) follows the conformal prediction previously established for the critical Ising chain [1]. This behaviour persists after introducing the NNN interaction, as Figure 8 shows for the Rényi-2 profiles at $p = 0.1$: once the finite-time corrections have been taken into account [23], the real part follows the chord profile of Eq. (8) and the imaginary part develops a plateau across the interior of the strip. Appendix G.1 gathers the profiles at all four couplings and the finite-time corrections that each of the two routes requires.

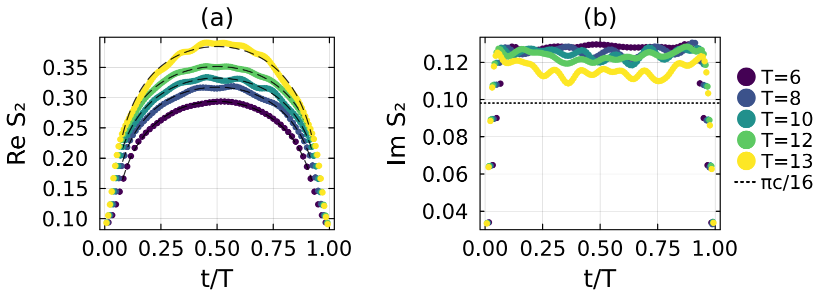

Figure 8: **Generalised temporal entropy at $p = 0.1$.** Each colour denotes one evolution time. (a) Real part as a function of t/T. The dashed curves show the conformal profile at $c = \frac{1}{2}$ with its finite-time correction, the two non-universal constants being fitted to each profile (Appendix G.1). (b) Imaginary part, which forms an approximately constant plateau in the interior of the strip. The plateau lies above the asymptotic value $\pi c/16$ (dotted line) and approaches it as T increases.

To extract the entropy-based central charge, we first average the imaginary part of the entropy over the four temporal bonds at the centre of the strip for each run. Residual lattice effects produce small oscillations across the otherwise approximately constant interior. This short average suppresses those oscillations while excluding the ends of the strip, where boundary effects dominate. We define the resulting average as the plateau value for that run. Repeating the calculation from independent random initialisations and taking the median over the physical runs gives one plateau value at each evolution time.

As the finite-time correction described in Section 2.2 decays, the plateau approaches the asymptotic value $\pi c/16$. At the integrable point, for example, it decreases from 0.132 at $T = 4$ to 0.108 at $T = 22$. We determine the central charge by fitting the predicted $T^{-1/2}$ correction [23] and extrapolating to $T \to \infty$. Appendix G.1 gives the fitting form and compares it with alternative corrections.

The reliability of this extrapolation is controlled by the length of the available time sequence. Over a short window, the predicted decay and a constant model can describe the data comparably well while giving different asymptotic values. The longer sequences at $p = 0$ and $p = 0.1$ therefore yield estimates consistent with the Ising value within their uncertainties. By contrast, the shorter sequences at $p = 0.3$ and $p = 0.5$ remain above it. The values and fitting windows are reported in Table 1. Each window ends when the ensembles of Section 6.3 cease to produce physical profiles, which occurs earlier as p increases (Appendix F.5). The quoted uncertainties measure the run-to-run variation of the plateau extrapolation, but do not quantify the remaining finite-time bias.

7.2 The spectral route

We compute the leading spectrum over the time windows listed in Table 1. Figure 9 shows that the trajectory of μ_0 remains close to a circle whose radius depends on the coupling: its phase winds with T, while its modulus varies by approximately 1% or less over each sequence, which is the signature of emergent dual unitarity described in Section 2.2 [1, 22]. The phase increment between successive times also remains nearly constant along the physical branch, so a persistent departure from this progression indicates that the branch is no longer resolved and determines the end of the fitting window.

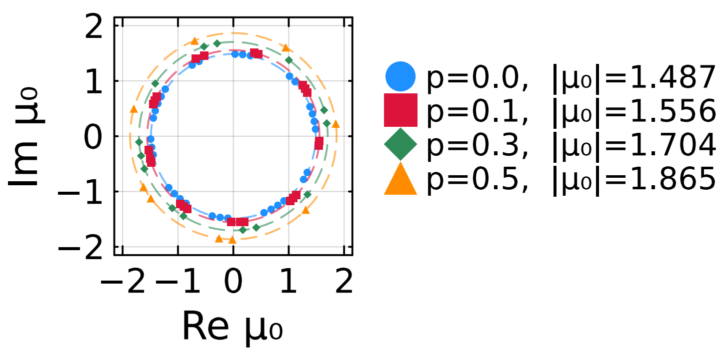

Figure 9: **Trajectory of the leading eigenvalue.** For each coupling, the dashed circle has the mean modulus of the corresponding sequence and is labelled by this value. As T increases, the phase winds while the modulus remains approximately constant.

We extract the central charge from the phase curvature using Eq. (6). Figure 10 shows the resulting fits.

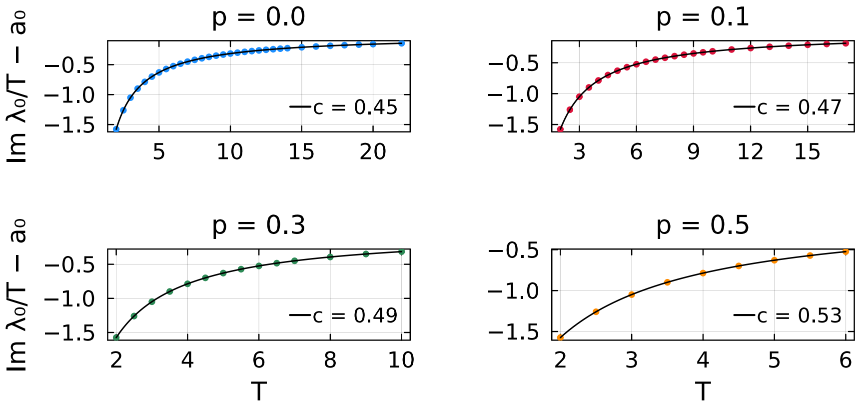

Figure 10: **Central charge from the phase of the leading eigenvalue.** Fits of Eq. (6) to $\mathrm{Im}\,\lambda_0/T$ at each coupling. The fitted constant a_0 is subtracted to isolate the time-dependent contribution, whose curvature determines the central charge reported in each legend.

The spectral estimates in Table 1 lie within approximately 10% of the Ising value at every coupling. At the integrable point, where the answer is known, the estimate is biased downwards to approximately 0.45. Two effects account for it, and both act in the same direction, which is why the estimates sit below $\frac{1}{2}$ rather than scattering around it.

The first is the fit itself. The production form truncates the expansion in Eq. (6) after the C/T^2 term, and the leading contribution it discards enters $\mathrm{Im}\,\lambda_0$ at order T^{-3}. At the integrable point, including that contribution raises the fitted coefficient C from 90% to 97% of its predicted value. The shorter interacting windows cannot constrain this additional parameter, so we use the same two-term form at every coupling and treat the remaining difference as a systematic bias.

The second is the regulator, which enters the analysis through the complex strip width $\beta = v(2\beta_0 + iT)$ in Eq. (2). Its effect can be measured directly: the fitted coefficient decreases approximately linearly with β_0, lies about 12% below the prediction at the production value $\beta_0 = 0.2$, and extrapolates to the predicted value as $\beta_0 \to 0$. That limit is not an accessible setting, because the cooling interval is required to produce the regularised conformal boundary state of the strip construction. The observed deficit is therefore consistent with the finite-time and finite-regulator effects described in Appendix G.1.

The equilibrium analysis of Section 4.1 gives comparable estimates of the central charge over the same coupling range, which provides an independent comparison between universal data obtained from the ground state and from real-time dynamics.

The sound velocity affects the spectral analysis in two ways. First, Eq. (7) gives phase gaps proportional to $1/(vT)$, so a larger velocity compresses the tower at fixed evolution time and requires finer phase resolution. Second, the conversion from the fitted curvature to the central charge also contains a factor of v, since $c = 24\,v\,|C|/\pi$. A fixed uncertainty in C therefore produces an uncertainty in c that increases with the velocity. Both effects reduce the spectral resolution at stronger coupling. Together with the worsening conditioning of the left and right eigenvectors (Appendix F.5), they may explain why the available windows shorten as p increases.

The velocity works the other way as well, however. Since it sets the length of the strip in conformal units, a larger v means the asymptotic regime is reached at a smaller lattice time: measured in the product $v_\infty T$, the four windows are far more alike than they appear, spanning 44, 45, 40 and 31 from $p = 0$ to $p = 0.5$. That is why a sequence ending at $T = 6$ still returns a sensible central charge at the strongest coupling.

On the other hand, the resolved boundary spectrum remains consistent with the Ising prediction after introducing the NNN interaction. Figure 11 shows the seven leading dimensions obtained with $k = 8$. The leading dimension, reported in Table 1, remains within 3% of the predicted value $\frac{1}{2}$ at every coupling.

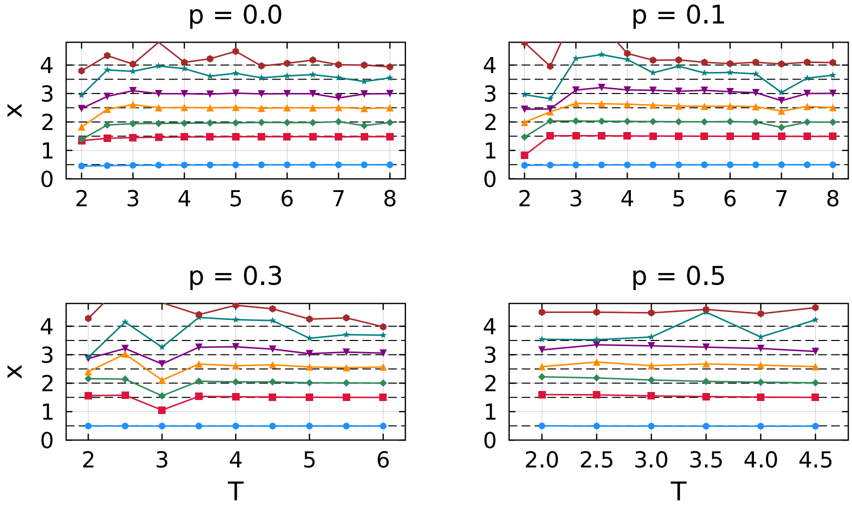

Figure 11: **Boundary dimensions as a function of evolution time.** The seven leading dimensions at each coupling are obtained from the phase gaps through Eq. (7), ordered by increasing phase distance from μ_0, and resolved with $k = 8$. The dashed lines mark the predicted values $\frac{1}{2}$, $\frac{3}{2}$, 2, $\frac{5}{2}$, 3, $\frac{7}{2}$, and 4. The free boundary condition selects two towers: the identity tower contains the integer values, while the energy tower contains the half-integer values (Appendix G.2).

The first five dimensions remain close to their predicted values throughout the coupling range. The sixth and seventh dimensions, which seem to drift slightly upwards compared with the predictions, are extracted from the highest states retained by the block and are therefore the least converged, as quantified in Appendix F.2. Their phase gaps are also the most difficult to resolve as the spectrum becomes more compressed, and at the stronger couplings the accessible window in T ends before these estimates can settle, so the conformal prediction cannot yet emerge clearly there.

8 Conclusions and Outlook

The question this thesis aimed to answer was whether the conformal signatures encoded in the Loschmidt echo persist when integrability is broken. We studied the self-dual ANNNI-type chain at criticality, mapped the echo to a CFT partition function on a strip, and extracted universal data from the transfer-matrix spectrum. For the range of couplings we analysed, $p \leq 0.5$, the results remain consistent with the Ising CFT wherever the numerical methods converge.

The dynamical observables exhibit the expected conformal behaviour. The leading eigenvalue retains an approximately constant modulus while its phase winds, as predicted by the emergent dual unitarity of Eq. (5). Once the finite-time corrections are accounted for, the phase curvature gives a central charge consistent with $c = \frac{1}{2}$, the real part of the generalised temporal entropy follows the predicted chord profile, and the imaginary part approaches $\pi c/16$ over the longer time windows. The phase gaps resolve the first seven boundary dimensions and, at sufficiently long times, recover the two conformal towers selected by the free boundary condition. Within the accessible coupling range, the NNN interaction therefore changes only the non-universal scales and does not alter the universal content.

The equilibrium calculations provide an independent identification of the universality class and determine the sound velocity required by the dynamical analysis. The DMRG estimates of the ground-state central charge remain consistent with the Ising value up to $p = 1.5$. Exact diagonalisation of finite rings gives the velocity at each coupling and shows that it increases substantially over the range considered.

The first numerical development of this work was a time-evolution MPO for the NNN interaction. A brick-wall Trotter circuit cannot assign this interaction to a single bond while preserving the one-site translation invariance required for transverse contraction. We instead exponentiated the FSM representation of the Hamiltonian directly. The resulting MPO implements the NNN terms while retaining translation invariance and is accurate to second order in the time step.

The second development was a block power method for resolving the subleading eigenvalues of the non-Hermitian transfer matrix (Appendix F). The leading eigenvalue and temporal entropies require only the dominant left and right eigenvectors, which we obtain with a single-vector iteration. Extracting the boundary spectrum instead requires a block of leading eigenvalues. Propagating and compressing this block is more expensive and restricts the spectral calculation to shorter evolution times.

The time windows in Table 1 are determined from numerical consistency checks, independently of their agreement with the conformal predictions. We require the entropy profiles to be reproducible across independent random seeds and the physical eigenvalue branch to evolve continuously between successive times. Additional controls vary the truncation scheme, bond dimension, singular-value cutoff, and time step, but none of them changes the reported values within the quoted precision.

We have not yet been able to identify the mechanism that limits these windows. The eigenvalues are more robust than the eigenvectors: $|\mu_0|$ remains reproducible across independent seeds throughout the usable range, whereas the entropies constructed from the corresponding eigenvectors cease to agree at earlier times. These disagreements appear sooner as p increases and coincide with growing ill-conditioning of the left and right eigenvectors. The controls in Appendix F.5 rule out several possible numerical causes but do not explain why the conditioning depends so strongly on the NNN coupling. The reported windows should therefore be understood as limits of the present numerical configuration, rather than fundamental limits of transverse contraction. Why the accessible window closes so quickly as p increases is the main question this work leaves open.

The block iteration is one part of our method with clear room for improvement. Most of its cost comes from propagating and compressing the block states, and its convergence test chases a tolerance that truncation makes unreachable, so runs end only after a fixed number of iterations without further improvement. A cheaper, better-conditioned implementation would extend the accessible windows and make the physical branch easier to track, which the non-normality of the transfer matrix complicates. Longer windows would let $c(p)$ and $x_1(p)$ be followed towards the boundary of the critical phase, where the equilibrium results remain consistent with Ising criticality but the present dynamical calculations are inconclusive.

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A Conformal-field-theory derivations

Section 2 relates the Loschmidt echo of a critical quench to the partition function of a CFT on a strip whose width is determined by the evolution time. The analysis uses two results derived here: the logarithmic coefficient of the generalised temporal entropy and the spectrum of the transfer matrix along the strip. We first obtain the entropy coefficient and chord variable from twist-operator correlation functions, including the factor of two associated with an edge-anchored bipartition. We then quantise the theory on the strip and continue its width to complex time, obtaining Eqs. (2), (6), and (7).

A.1 Conformal maps, the two-point function and the entropy coefficients

In two dimensions, we write the Euclidean coordinates as $z = x + i\tau$ and $\bar{z} = x - i\tau$. A conformal map is a transformation $z \to w(z)$ depending on z alone, together with an independent transformation $\bar{z} \to \bar{w}(\bar{z})$ of the conjugate coordinate. The holomorphic and antiholomorphic dependences therefore transform separately, and we call each one a chiral sector; in real time they carry the left- and right-moving excitations. A primary operator ϕ is one that transforms under such a map with the Jacobian alone, without the derivative terms a general operator acquires, and its conformal weights $(h, \bar{h})$ set the exponent contributed by each sector. Their sum,

$$x = h + \bar{h} \tag{A.1}$$

is the scaling dimension used in the main text, and their difference $s = h - \bar{h}$ is the spin, which measures the response to a rotation about the operator. The twist operators used below sit at branch points of the replicated surface, which are invariant under such a rotation, so $h = \bar{h}$ and $s = 0$. On the plane, the two-point function has the power-law form

$$\langle \phi(z_1)\phi(z_2)\rangle = \frac{1}{(z_1 - z_2)^{2h}\,(\bar{z}_1 - \bar{z}_2)^{2\bar{h}}}. \tag{A.2}$$

Under a conformal map $z \to w$, a primary field transforms with the corresponding Jacobian. Equation (A.2) therefore determines its correlator on any geometry conformally related to the plane [29]. The logarithmic map $w = (\beta/\pi)\log z$ maps the upper half-plane to a strip of width β and the full plane to a cylinder of circumference 2β. Here β is a length, so on the lattice it carries the sound velocity through $\beta = v(2\beta_0 + iT)$, and v does not appear separately in the map. The transformed correlator is [1, 32]

$$\langle \phi(\omega_1)\phi(\omega_2)\rangle \;\propto\; \left[\sinh \tfrac{\pi(\omega_1-\omega_2)}{2\beta}\right]^{-2h} \left[\sinh \tfrac{\pi(\bar{\omega}_1-\bar{\omega}_2)}{2\beta}\right]^{-2\bar{h}}. \tag{A.3}$$

The separation on the plane is replaced by the chord distance $(2\beta/\pi)\sinh[\pi(\omega_1 - \omega_2)/2\beta]$ on the cylinder or strip. Analytic continuation of the width to real time replaces the hyperbolic sine by a sine. This continued chord distance gives the variable W in Eq. (4).

The entropy coefficients follow from the same correlation function. The Rényi entropy of a region $\mathcal{A}$ is

$$S_n = \frac{1}{1-n}\log \operatorname{Tr}\rho_{\mathcal{A}}^n, \tag{A.4}$$

so the quantity to be computed is the moment $\operatorname{Tr}\rho_{\mathcal{A}}^n$, the trace of the nth power of the reduced density matrix. For integer n this moment is itself a partition function: each factor of $\rho_{\mathcal{A}}$ is written as a path integral, and taking the trace joins n copies of the system cyclically along $\mathcal{A}$ [32, 41]. Analytic continuation to $n \to 1$ gives the von Neumann entropy. Each endpoint of $\mathcal{A}$ in the interior is a branch point of the replicated surface and is represented by a primary twist operator of the n-fold replicated theory with scaling dimension [32]

$$x_n = \frac{c}{12}\Big(n - \frac{1}{n}\Big), \tag{A.5}$$

whose weights are $h_n = \bar{h}_n = x_n/2$ in the notation of Eq. (A.1).

For an interval of length ℓ with both endpoints in the interior, the moments are determined by the two-point function of two twist operators. Equation (A.2) gives $\operatorname{Tr}\rho_{\mathcal{A}}^n \sim \ell^{-2x_n}$ and therefore

$$S_n = \frac{c}{6}\Big(1 + \frac{1}{n}\Big)\log\ell + \text{const}, \qquad S_{n\to 1} = \frac{c}{3}\log\ell + \text{const}. \tag{A.6}$$

The temporal bipartitions used in this thesis terminate at an edge of the strip and have only one endpoint in the interior. Their moments therefore involve the one-point function of a single twist operator. A conformal boundary relates the two chiral sectors, so the antiholomorphic part of an operator can be replaced by a holomorphic one at its mirror image [31]. The one-point function is then evaluated as a correlator between the operator and that image, separated by twice the distance to the boundary [32]. Because a single operator is inserted rather than two, the exponent is x_n instead of $2x_n$ and the logarithmic coefficient is halved:

$$S_n = \frac{c}{12}\Big(1 + \frac{1}{n}\Big)\log\ell + \text{const}. \tag{A.7}$$

On a strip of finite extent ℓ, the distance between an operator at x and its image is $(2\ell/\pi)\sin(\pi x/\ell)$, which is the argument of the chord variable $W(x,\ell)$ in Eq. (4), so the edge-anchored Rényi entropy is

$$S_n = \frac{c}{12}\Big(1 + \frac{1}{n}\Big)\,W(x,\ell) + s_0, \tag{A.8}$$

where s_0 is non-universal. For an open chain the same result is written with ℓ/π rather than $2\ell/\pi$ [32]. This difference contributes only an additive constant and is absorbed into s_0.

To obtain Eq. (3), we continue the strip extent to real time by setting $\ell \to i\,vT$ and $x \to i\,vt$. The ratio x/ℓ remains real, while the prefactor inside the logarithm acquires a factor of i. Using $\log i = i\pi/2$ gives

$$S_n^{\text{gen}}(t) = s_0 + \frac{c}{12}\Big(1 + \frac{1}{n}\Big)\Big[\,W(t,T) + \frac{i\pi}{2}\,\Big]. \tag{A.9}$$

This reproduces Eq. (3). The real part contains the chord profile, while the imaginary part is the constant $\frac{\pi c}{24}(1 + \frac{1}{n})$, equal to $\pi c/16$ at $n = 2$. The velocity contributes only an additional constant inside the logarithm and is absorbed into s_0.

A.2 The transfer-matrix spectrum from the stress tensor

The stress-energy tensor determines the response of the theory to coordinate transformations. In two dimensions, its modes L_n and $\bar{L}_n$ generate conformal transformations in the two chiral sectors and satisfy one copy of the Virasoro algebra each, whose central term is proportional to c [29]. The mode L_0 generates scale transformations, and its eigenvalues on primary operators are their conformal weights. Its spectrum therefore encodes the operator content.

Conformal boundary conditions at the two edges of the strip identify the chiral sectors. Extending the stress tensor into the lower half-plane through $\bar{T}(z) = T(z)$ leaves a single Virasoro algebra [31]. The dimensions below are therefore the weights h of boundary operators, rather than the bulk dimensions $h + \bar{h}$ defined in Eq. (A.1); we continue to write them x_i.

The generator of translations along the strip is obtained from the stress tensor. Since the stress tensor is not a primary field, a conformal map $z \to w$ introduces an inhomogeneous Schwarzian term proportional to the central charge [29]:

$$T(w) = \Big(\frac{\mathrm{d}z}{\mathrm{d}w}\Big)^2 T(z) + \frac{c}{12}\,\{z; w\}. \tag{A.10}$$

The map $w = (\beta/\pi)\log z$ from Appendix A.1 takes the upper half-plane to a strip of width β and has Schwarzian derivative $\{z; w\} = -\frac{1}{2}(\pi/\beta)^2$. Since $\langle T\rangle$ vanishes on the half-plane, its expectation value on the strip is $\langle T(w)\rangle = -\pi^2 c/(24\beta^2)$. Integrating across the strip gives a universal contribution $-\pi c/(24\beta)$ to the free energy per unit length [30]. Including the excitation levels generated by dilatations, the translation generator is [35]

$$\hat{H} = \frac{\pi}{\beta}\Big(L_0 - \frac{c}{24}\Big). \tag{A.11}$$

It contains both universal quantities of interest: the Casimir term $-c/24$ determines the central-charge contribution, while the spectrum of L_0 gives the boundary scaling dimensions. The transfer matrix advances the network by one spatial column, and $\hat{H}$ generates translations in that direction. Hence $\mathcal{E} = e^{-a\hat{H}}$, where a is the column spacing [1]. Substituting Eq. (A.11) and setting $a = 1$ in lattice units reproduces Eq. (2), with $g = a/\beta$ and eigenvalues

$$\mu_i = \exp\Big[-\pi g\Big(x_i - \frac{c}{24}\Big)\Big]. \tag{A.12}$$

The identity sector has $x_0 = 0$ and gives the dominant eigenvalue. Differences between logarithmic eigenvalues therefore cancel the Casimir contribution and isolate the boundary dimensions.

The continuum expression must be supplemented by two non-universal lattice contributions. The free energy per unit length contains a bulk term and a contribution from the two boundaries [30], giving the level energies

$$E_i = f\beta + f_s + \frac{\pi}{\beta}\Big(x_i - \frac{c}{24}\Big), \tag{A.13}$$

where f and f_s depend on the microscopic model, Trotter step, and regulator. Since both terms are common to every level, they multiply all eigenvalues of $\mathcal{E}$ by the same factor. They therefore cancel in the phase differences used to extract the dimensions. The leading phase is instead fitted as a series in $1/T$ with a term linear in T. Since β is proportional to T, the bulk term $f\beta$ contributes a term linear in T, while f_s is real and shifts only the modulus. Neither reaches the $1/T$ coefficient, which is the one carrying the central charge, and neither contributes a constant to the phase either: $f\beta$ is linear in T and f_s is real. The constant term in $\operatorname{Im}\lambda_0$ is therefore the branch offset alone.

The effective strip width contains both imaginary- and real-time contributions. The regulator supplies imaginary-time intervals at the two boundaries, the quench supplies the intervening real-time interval, and the velocity converts these durations into lengths:

$$\beta = v\,(2\beta_0 + i\,T), \qquad \frac{1}{\beta} \simeq \frac{1}{v}\Big(\frac{2\beta_0}{T^2} - \frac{i}{T}\Big), \tag{A.14}$$

where the second expression is expanded to first order in β_0/T. Substitution into Eq. (2) gives

$$\log \mu_i \;\simeq\; \frac{i\pi}{v\,T}\Big(x_i - \frac{c}{24}\Big) \;-\; \frac{2\pi\beta_0}{v\,T^2}\Big(x_i - \frac{c}{24}\Big). \tag{A.15}$$

The universal contributions appear in the phases at order $1/T$, whereas the regulator changes the moduli at order β_0/T^2. The eigenvalues follow from the level energies as $\log \mu_i = -E_i$ at $a = 1$. Restoring the non-universal terms of Eq. (A.13), setting $x_0 = 0$ for the identity, and taking the imaginary part of $\lambda_0 = \log(-\mu_0)$ gives

$$\operatorname{Im} \lambda_0 = A\,T + B + \frac{C}{T} + \mathcal{O}(T^{-3}), \qquad A = -fv, \quad B = -\pi, \quad C = -\frac{\pi c}{24\,v}. \tag{A.16}$$

This is the form of Eq. (6) fitted in Section 7. The extensive term $f\beta$ fixes A and the factor -1 in the logarithm fixes B, while the coefficient C contains the central charge. Taking the difference between two logarithmic eigenvalues removes the common terms, including the Casimir contribution, and gives Eq. (7). The same expansion gives the emergent dual unitarity of Eq. (5). Writing the width as a geometric series,

$$\frac{1}{\beta} = -\frac{i}{v\,T} \sum_{k \geq 0} \Big(\frac{2i\beta_0}{T}\Big)^k, \tag{A.17}$$

the prefactor is imaginary and each further factor multiplies by i, so the terms with even k are imaginary and those with odd k are real. Only the real part contributes to the moduli, which therefore receive even powers of $1/T$ alone, each carrying at least one power of β_0. There is consequently no $1/T$ term: that contribution is purely imaginary and goes entirely into the phase. Keeping the leading real term,

$$|\mu_i| = \exp\Big[-\frac{2\pi\beta_0}{v\,T^2}\Big(x_i - \frac{c}{24}\Big)\Big]. \tag{A.18}$$

The Casimir term is common to every level, as are the non-universal contributions of Eq. (A.13), since $\operatorname{Re}(f\beta) = 2fv\beta_0$ does not depend on the level. All of them cancel in the ratio, leaving

$$\frac{|\mu_i|}{|\mu_0|} = \exp\Big[-\frac{2\pi\beta_0\,(x_i - x_0)}{v\,T^2}\Big] \simeq 1 - \frac{2\pi\beta_0\,(x_i - x_0)}{v\,T^2}, \tag{A.19}$$

which is Eq. (5) with $b_i = 2\pi\beta_0(x_i - x_0)/v$. The ratio therefore approaches unity both as T grows and as the regulator is removed at fixed T, and the emergent dual unitarity tested in Section 7 follows.

B Truncation schemes

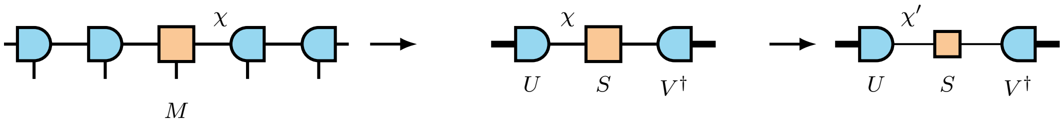
Truncation keeps the bond dimension bounded during transverse contraction. We now examine the two schemes introduced in Section 3.4, based on the RDM and the RTM. We first explain why Schmidt-value truncation is optimal for a single state. We then use gauge transformations to distinguish physical spectra from representation-dependent ones and identify the approximations involved in the RTM scheme.

Consider a cut that divides the chain into a block $\mathcal{A}$ and its complement $\mathcal{B}$ [3, 40]. Grouping the physical indices on each side into a composite index represents the wavefunction as a matrix. Its Singular Value Decomposition (SVD) gives the Schmidt decomposition

$$|\psi\rangle = \sum_{\alpha=1}^{\chi} s_\alpha \;|\alpha\rangle_{\mathcal{A}} \otimes |\alpha\rangle_{\mathcal{B}}\,, \qquad s_\alpha \geq 0, \quad \sum_\alpha s_\alpha^2 = 1, \tag{B.1}$$

where the states on each side form orthonormal bases. In mixed canonical form, the Schmidt coefficients appear as a diagonal matrix on the cut bond. The corresponding RDM is diagonal in the same basis, with eigenvalues $p_\alpha = s_\alpha^2$ that determine the entanglement across the cut. Reducing the bond dimension from χ to $\chi' < \chi$, as shown in Figure 12, retains the χ' largest Schmidt coefficients. The Eckart–Young–Mirsky theorem [53, 54] establishes that this is the closest state to $|\psi\rangle$ among all states of Schmidt rank at most χ'. Its squared norm error is the discarded weight,

$$\varepsilon = \| \, |\psi\rangle - \left|\tilde{\psi}\right\rangle \|_2^2 = \sum_{\alpha > \chi'} s_\alpha^2. \tag{B.2}$$



(a) canonical form

(b) we group the legs to form matrices

(c) we truncate

Figure 12: **Schmidt truncation at a bond.** (a) Chain in mixed canonical form, with the orthogonality centre M at the cut. (b) Grouping the indices on either side into composite indices (thick lines) gives the matrix decomposition $M = USV^\dagger$. (c) Truncated decomposition, with the internal dimension reduced from χ to χ'.

B.1 The RDM scheme

In canonical form, the RDM scheme obtains this truncation from a local tensor, as shown in Figure 13.

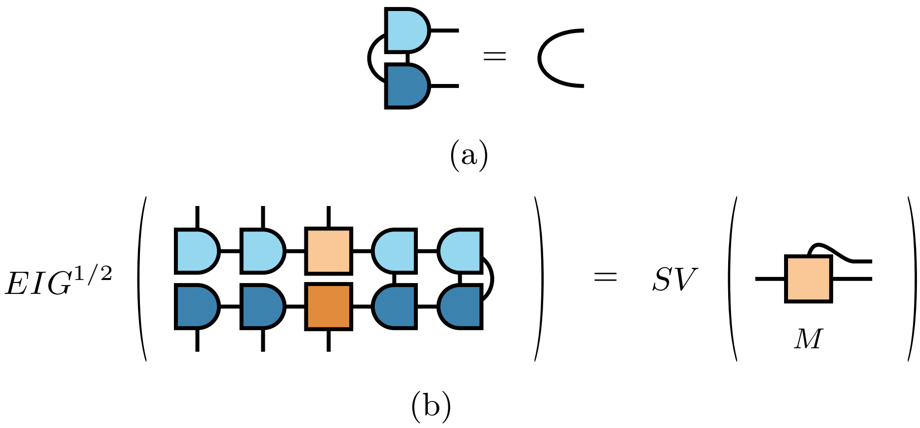


Figure 13: **Local RDM truncation in canonical form.** (a) Contracting a canonical tensor with its conjugate over the physical index and one virtual index gives the identity on the remaining index. Lighter and darker shades distinguish the tensor and its conjugate. (b) Reduced density matrix at a cut, with the left block open and the right block traced. The surrounding isometries contract to identities, so the eigenvalues of the full RDM are the squared singular values of the centre tensor M.

The truncation data can therefore be extracted from a local $\chi \times \chi$ environment matrix M_A rather than from the exponentially large global object. For a bipartite cut,

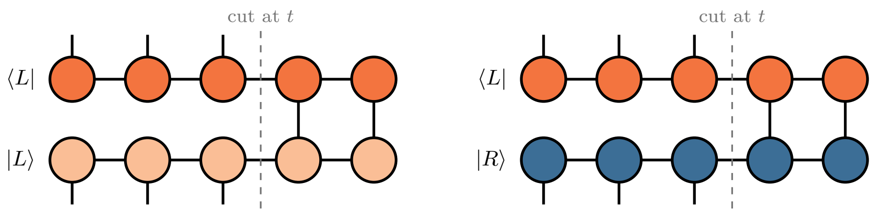
$$\rho_B = U_B \, M_A \, U_B^\dagger, \tag{B.3}$$

where U_B is an isometry because it is constructed from left-orthogonal tensors, so $U_B^\dagger U_B = \mathbb{1}$. Equation (B.3) therefore preserves the nonzero spectrum, $\mathrm{eig}(\rho_B) = \mathrm{eig}(M_A)$ up to additional zeros. The local environment carries the same eigenvalues as the full RDM.

This construction relies on the RDM being Hermitian and positive. Its eigenvalues are the squared Schmidt coefficients and can therefore be ordered unambiguously, while the Eckart–Young–Mirsky theorem provides a variational error bound. Neither property extends directly to a non-Hermitian transition matrix.

B.2 The RTM scheme

In the transverse picture, the transfer matrix is non-Hermitian and its dominant left and right fixed points are distinct. The corresponding truncation object is the RTM in Eq. (11), constructed from both states. Its eigenvalues can be complex and do not have the Schmidt interpretation described above. Moreover, a gauge transformation can redistribute weight between components that contribute to $\langle L|R\rangle$ and those that do not [39]; the magnitude of a component in either state separately is therefore not a gauge-invariant measure of its contribution to the overlap. Figure 14 compares the RDM and RTM at the same temporal cut.



(a) $\rho_L = \mathrm{Tr}_{T-t}\,|L\rangle\langle L|$ (b) $\mathcal{T}_t = \mathrm{Tr}_{T-t}\,|R\rangle\langle L|\,/\,\langle L|R\rangle$

Figure 14: **Possible truncation objects at a temporal cut.** Sites after the cut are traced over, while the physical legs before the cut remain open and form the matrix indices. (a) The RDM pairs a temporal state with its conjugate and is Hermitian and positive. (b) The RTM pairs the distinct fixed points $|R\rangle$ and $\langle L|$; it is generally non-Hermitian and can have complex eigenvalues.

The tensor representation of either object is not unique. Let $A^{[n]}$ and $A^{[n+1]}$ be the MPS tensors on two sites joined by a virtual bond. We may insert $\mathbb{1} = XX^{-1}$ on this bond and absorb one factor into each tensor,

$$A^{[n]} \;\to\; A^{[n]}X, \qquad A^{[n+1]} \;\to\; X^{-1}A^{[n+1]}, \tag{B.4}$$

without changing the represented state, because the two factors cancel when the bond is contracted [3, 4]. The physical overlap is invariant under this transformation,

$$\langle L|R\rangle \;=\; \langle L|\,XX^{-1}\,|R\rangle \;=\; \left\langle \tilde{L} \middle| \tilde{R} \right\rangle, \tag{B.5}$$

but the spectra used for truncation need not be. Their transformation depends on whether the state across the cut is paired with its conjugate or with an independent left state.

The RDM pairs the state with its conjugate. Conjugating the first transformation in Eq. (B.4) gives $A^{[n]*} \to A^{[n]*}X^*$. Thus, where the ket carries X, the bra carries $X^\dagger$, and

$$\rho_L \;\to\; X^\dagger \rho_L X. \tag{B.6}$$

The factors do not cancel unless X is unitary, because $X^\dagger X \neq \mathbb{1}$ in general. Equation (B.6) is a congruence transformation and does not preserve the eigenvalues. A gauge transformation can therefore move components that contribute to the overlap into the tail of the isolated RDM spectrum, where they may be discarded by truncation.

The RTM instead pairs $|R\rangle$ with the independent state $\langle L|$. The left tensors absorb X^{-1} on the other side of the same bond, following the second transformation in Eq. (B.4). Consequently,

$$\mathcal{T}_t \;\rightarrow\; X^{-1}\mathcal{T}_t X. \tag{B.7}$$

The factors now cancel because $X^{-1}X = \mathbb{1}$. Equation (B.7) is a similarity transformation for any invertible X and therefore preserves the eigenvalues. Figure 15 shows the corresponding network contractions for both schemes.

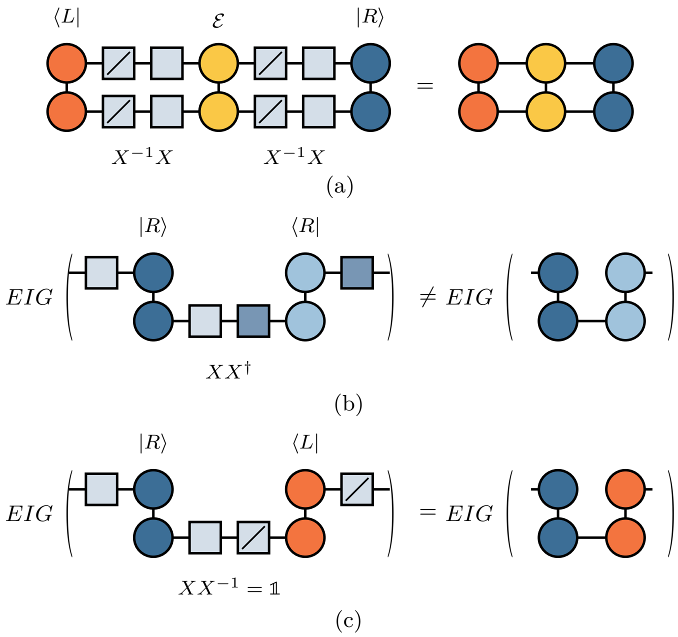


Figure 15: **Gauge transformation of the RDM and RTM.** (a) Inserting $X^{-1}X$ on a virtual bond changes the tensors but not the full contraction. In (b) and (c), the lower time site is contracted while the upper site remains open. (b) For the RDM, the gauge factors combine as $XX^\dagger \neq \mathbb{1}$ and produce the congruence $X\rho X^\dagger$, which changes the spectrum. (c) For the RTM, they combine as $XX^{-1} = \mathbb{1}$ and produce the similarity transformation $X\mathcal{T}_t X^{-1}$, which preserves the spectrum.

The spectrum of a single temporal-state RDM therefore depends on the chosen representation, whereas the RTM spectrum is invariant for the pair $\langle L|, |R\rangle$. Fixing a canonical gauge makes the RDM truncation well defined, allowing it to be used as a complementary numerical scheme despite this representation dependence.

On the other hand, the local reduction of the RTM differs significantly from that of the RDM. For two distinct states, the construction leading to Eq. (B.3) instead gives

$$\tau_B = U_B\, M_A\, V_B, \qquad U_B \neq V_B^\dagger, \tag{B.8}$$

which is not a similarity transformation. Its eigenvalues are therefore not preserved: $\mathrm{eig}(\tau_B) \neq \mathrm{eig}(M_A)$. However, U_B and V_B remain isometries and preserve singular values,

$$\mathrm{sv}(\tau_B) = \mathrm{sv}(M_A). \tag{B.9}$$

Thus, the local environment and the full RTM have the same singular values. These are the invariant quantities available from the local reduction and provide the truncation criterion we use in practice.

This local reduction requires a canonical form adapted to the left–right pair. As shown in Figure 16, the canonical condition is imposed on the pair $(\langle L|, |R\rangle)$ rather than on a state and its conjugate. Tensors away from the cut then contract to identities, leaving a local decomposition with the singular values of the full RTM. The two schemes consequently have the same local structure and comparable cost at fixed bond dimension.

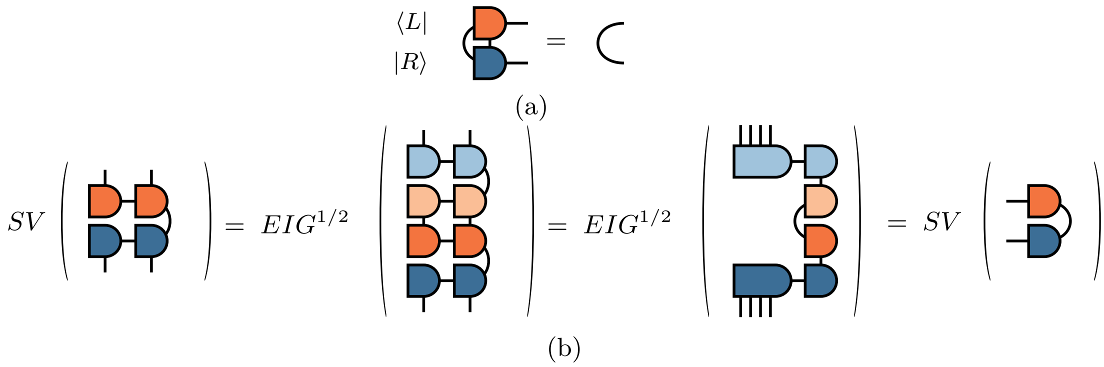


Figure 16: **Local RTM truncation in generalised canonical form.** (a) The contraction pairs a tensor of $\langle L|$ with the corresponding tensor of $|R\rangle$ rather than with its conjugate. (b) Imposing this condition along the chain reduces the environment to the tensors at the cut, whose singular values equal those of the RTM. The construction remains local but no longer has a variational error bound.

The potential reduction in bond dimension arises because the RTM measures relevance through the left–right overlap. The method uses a biorthogonal decomposition of $|R\rangle\langle L|$ and retains the components with largest-magnitude eigenvalues [18]. For the transfer matrices considered here, this can reach a given accuracy with a smaller bond dimension than the RDM scheme. It is the rank of the RTM itself, rather than the temporal entropy of either boundary state alone, that dictates the bond dimension the calculation requires [18].

The local truncation criterion is expressed through singular values, whereas the physical overlap is determined by the trace of $\mathcal{T}_t$. These quantities are not equivalent. Singular values satisfy $\mathrm{sv}(M) = \sqrt{\mathrm{eig}(MM^\dagger)}$ and coincide with the eigenvalues for a Hermitian positive RDM, but not for a non-Hermitian RTM. The trace $\mathrm{Tr}\,\mathcal{T}_t = \sum_i \lambda_i$ is a signed sum and is not controlled directly by singular-value magnitudes. For example, eigenvalues $+1$ and -1 both have unit magnitude but cancel in the trace.

The RTM truncation therefore lacks the variational bound available for an RDM, and the same criterion need not be reliable for a generic non-Hermitian matrix, whose eigenvalues can respond singularly to perturbations when the matrix is close to defective [55]. The error in the overlap can nevertheless be bounded: if the truncation changes the left and right temporal states by at most ϵ_L and ϵ_R in norm, then the error in $\langle L|R\rangle$ is at most $\epsilon_L + \epsilon_R + \epsilon_L\epsilon_R$ [16]. The RTM truncation targets contributions to the overlap rather than minimising these norm errors, so the discarded singular values cannot be used directly to evaluate this bound. Cancellation of eigenvalues in the trace would require retained and discarded components of comparable size, which the quickly decaying spectra of the model studied here at the critical point do not provide. The different guarantees are reflected in the convergence: the RDM scheme generally converges smoothly, whereas the RTM result can vary irregularly; Figure 21 in Appendix F.5 shows the two behaviours at a single production point.

Both schemes make one further approximation. At each stage of the sweep, the truncation uses the current local environment without including the uncontracted remainder of the network. Translation invariance reduces the resulting error because every subsequent bulk column applies the same operator and can regenerate discarded components of the leading subspace.

C Why the NNN coupling breaks integrability

Mapping a one-dimensional spin chain to fermions provides a direct test of whether it reduces to free fermions. Under the JW transformation, a quadratic fermionic Hamiltonian can be reduced to independent normal modes and diagonalised exactly. Terms of fourth order instead couple these modes and remove this free-fermion integrability. We apply this criterion to Eq. (13).

The mapping must account for the different algebras of spins and fermions: Pauli operators on different sites commute, whereas fermionic operators anticommute. The JW transformation achieves this through a parity string over all sites to the left of the operator. In the convention of Eq. (13), with the transverse field along σ^x and the order parameter along σ^z, it is [37, 56]

$$\sigma_j^x = 1 - 2n_j, \qquad \sigma_j^z = (c_j + c_j^\dagger)\, K_j, \qquad K_j = \prod_{k<j} (1 - 2n_k), \tag{C.1}$$

where c_j annihilates a spinless fermion, $n_j = c_j^\dagger c_j$, and K_j measures the fermion parity to the left of site j.

Two identities determine how these strings cancel. Each parity factor squares to the identity, $(1-2n_k)^2 = 1$, and hence $K_j^2 = 1$. Moreover, $1 - 2n_k = (-1)^{n_k}$ is even in the fermionic operators and commutes with operators on other sites, while on the same site

$$(c + c^\dagger)(1 - 2n) = c^\dagger - c. \tag{C.2}$$

The transverse field maps directly to the chemical-potential term $\sigma_j^x = 1 - 2c_j^\dagger c_j$. For the NN interaction, using $K_{j+1} = K_j(1 - 2n_j)$ gives

$$\begin{aligned}
\sigma_j^z \sigma_{j+1}^z &= (c_j + c_j^\dagger) K_j \,(c_{j+1} + c_{j+1}^\dagger) K_j (1 - 2n_j) \\
&= (c_j + c_j^\dagger)(c_{j+1} + c_{j+1}^\dagger)\, K_j^2\, (1 - 2n_j) \\
&= (c_j + c_j^\dagger)(1 - 2n_j)\,(c_{j+1} + c_{j+1}^\dagger) \\
&= (c_j^\dagger - c_j)(c_{j+1} + c_{j+1}^\dagger).
\end{aligned} \tag{C.3}$$

The string commutes with the operators on site $j+1$ because it is even in the fermions. The remaining factor $(1-2n_j)$ is then combined with the operator on site j using Eq. (C.2). No parity factor survives, so the interaction is quadratic. Together with the field term, this makes the $p = 0$ chain a free-fermion model that can be diagonalised exactly by a Bogoliubov transformation [45]. At criticality, its continuum limit is the free Majorana CFT with $c = \frac{1}{2}$ [29].

For the NNN interaction, the parity string crosses the intervening site. Since $K_{i+2} = K_i(1 - 2n_i)(1 - 2n_{i+1})$, the cancellation leaves two parity factors:

$$\begin{aligned}
\sigma_i^z \sigma_{i+2}^z &= (c_i + c_i^\dagger) K_i \,(c_{i+2} + c_{i+2}^\dagger) K_i (1 - 2n_i)(1 - 2n_{i+1}) \\
&= (c_i + c_i^\dagger)(1 - 2n_i)\,(1 - 2n_{i+1})\,(c_{i+2} + c_{i+2}^\dagger) \\
&= (c_i^\dagger - c_i)\,(1 - 2n_{i+1})\,(c_{i+2} + c_{i+2}^\dagger).
\end{aligned} \tag{C.4}$$

The factor on site i is absorbed as in the NN case, but the factor on site $i+1$ remains because the bond contains no spin operator there. The hopping and pairing between sites i and $i+2$ therefore depend on the occupation of the intervening site.

Expanding the remaining parity factor separates the quadratic and quartic contributions:

$$\sigma_i^z \sigma_{i+2}^z = (c_i^\dagger - c_i)(c_{i+2} + c_{i+2}^\dagger) - 2\,(c_i^\dagger - c_i)\,c_{i+1}^\dagger c_{i+1}\,(c_{i+2} + c_{i+2}^\dagger). \tag{C.5}$$

The second term contains four fermionic operators. The self-dual interaction likewise contains a density–density contribution,

$$\sigma_j^x \sigma_{j+1}^x = 1 - 2n_j - 2n_{j+1} + 4\,c_j^\dagger c_j c_{j+1}^\dagger c_{j+1}. \tag{C.6}$$

Both terms proportional to p therefore contain four-fermion interactions. These interactions couple the normal modes of the $p = 0$ chain, so Wick's theorem no longer reduces correlation functions to products of two-point functions [57], and no single-particle transformation diagonalises the Hamiltonian, since a linear transformation of the modes preserves the order of the interaction terms. Thus $p \neq 0$ removes the quadratic free-fermion structure responsible for integrability at $p = 0$.

D Equilibrium measurements

The dynamical analysis in Section 7 requires two equilibrium quantities. The equilibrium central charge provides an independent reference for the dynamical estimate, while the sound velocity converts transfer-matrix phase gaps into scaling dimensions. We determine both quantities here.

D.1 Finite-size scaling of the mid-chain entropy

Section 4.1 extracts the central charge from the entanglement profile at a fixed system size. We obtain an independent estimate from the size dependence of the entropy across the mid-chain cut. At this cut, the chord variable is maximal, $W(L/2, L) = \ln(2L/\pi)$, and Eq. (14) becomes

$$S_{\mathrm{vN}}(L/2) = \frac{c}{6}\ln L + k'. \tag{D.1}$$

A linear fit against $\ln L$ therefore has slope $c/6$. For $N = 40$–200, the uncorrected fit gives $c = 0.511$ at $p = 0.1$ and $c = 0.514$ at $p = 0.5$. The same ground states give the Rényi-2 entropy used in the temporal calculation, whose logarithmic coefficient is $c/8$. The corresponding uncorrected estimates, $c = 0.584$ and 0.588, lie appreciably above the Ising value. Figure 17(a) shows both estimators.

This discrepancy is caused by finite-size corrections associated with the conical singularity of the replica geometry. Operators that are relevant at this singularity produce corrections proportional to $\ell^{-x/n}$ for an interval anchored at the boundary, where x is the scaling dimension of that operator [42]. The mid-chain cut of an open chain has this geometry, so we fit

$$S_n(L/2) = \frac{c}{12}\Big(1 + \frac{1}{n}\Big)\ln L + k_n + A_n L^{-x/n}. \tag{D.2}$$

with A_n a non-universal amplitude. Which operator appears at the singularity is not known in advance, so we determine its dimension from the data: fits with x free return values close to 1 at both n, although a three-parameter fit to nine sizes does not determine it sharply. We therefore fix $x = 1$. The corrected estimates are $c = 0.500$ at both couplings for the von Neumann entropy and $c = 0.508$ and 0.505 for the Rényi-2 entropy, and the largest residual falls by roughly two orders of magnitude in every case. Figure 17(b) shows the corrections that the fits absorb.

The factor x/n explains why the uncorrected estimates behave differently. For the same x, the correction to the von Neumann entropy decays as L^{-x}, whereas that of the Rényi-2 entropy decays as $L^{-x/2}$. The former is therefore already within 3% of $\frac{1}{2}$, while the latter requires the correction term. The analogous temporal correction scales as $\Delta S_n \propto T^{-1/n}$ [23] and is included in Section 7.1.

The finite-size calculations use fifteen DMRG sweeps, a bond-dimension ramp up to 400, and a truncation cutoff of 10^{-10}, beginning with noise-assisted sweeps. The $N = 400$ chord fit in Section 4.1 instead uses fifty sweeps, a maximum bond dimension of 1000, and a cutoff of 10^{-12}.

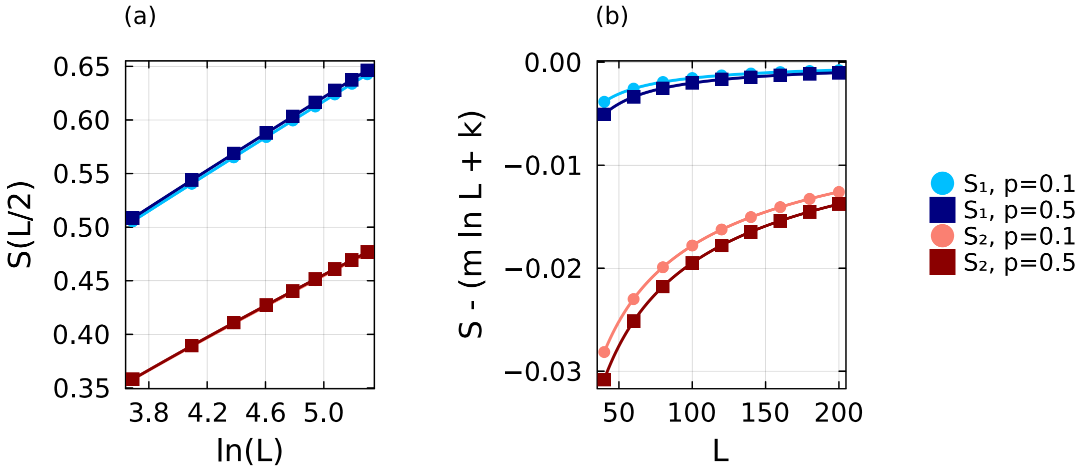


Figure 17: **Finite-size scaling of the mid-chain entropy.** (a) The von Neumann entropy S_1 and Rényi-2 entropy S_2 against $\ln L$ at $p = 0.1$ and $p = 0.5$, with the fits of Eq. (D.2). The corrected estimates are $c = 0.500$ at both couplings from S_1, and $c = 0.508$ and 0.505 from S_2. The correction changes the curves at the 10^{-3} level and is not visible on this scale. (b) The same data after subtracting the fitted logarithmic contribution, which isolates the finite-size correction.

D.2 The sound velocity from the equilibrium dispersion

We obtain the sound velocities used in Section 4.2 from the low-momentum dispersion, $\omega = v|k|$. On a periodic chain of N sites, the allowed momenta are $k_m = 2\pi m/N$, so a finite-size estimate follows from the lowest excitations at $m = 0$ and $m = 1$. These states must be identified within the same symmetry sector before their energy difference can be interpreted as a dispersion.

We therefore diagonalise the Hamiltonian in Eq. (13) at $\lambda = 1$ together with its translation symmetry. The one-site translation operator T has eigenvalues $e^{2\pi i m/N}$ and determines the momentum index m. It commutes with the Hamiltonian, so every level carries a definite momentum.

Degenerate eigenspaces require additional care because numerical diagonalisation returns arbitrary linear combinations of states with the same energy. We group levels whose energies differ by less than 10^{-8} and, within each cluster, diagonalise T, which restores the labels. A shift by N sites returns the ring to itself, so $T^N = \mathbb{1}$ and the eigenvalues of T are the Nth roots of unity, $e^{2\pi i m/N}$ with m an integer. An eigenstate therefore has

$\langle T\rangle = e^{2\pi i m/N}$, and its phase gives m directly. The computed phase is not exactly a multiple of $2\pi/N$, so we round it to the integer it must be, and assign

$$m = \mathrm{round}\left[\frac{N}{2\pi}\arg\langle T\rangle\right]. \tag{D.3}$$

With arg taken in $(-\pi, \pi]$ the index is signed, and the momentum is $k_m = 2\pi m/N$. Only the ten lowest levels are computed.

The finite-size velocity is then

$$v(N) = \frac{E_1(2\pi/N) - E_1(0)}{2\pi/N} = \frac{N}{2\pi}\left[E_1(2\pi/N) - E_1(0)\right]. \tag{D.4}$$

This is the discrete form of the low-momentum slope $\mathrm{d}\omega/\mathrm{d}k$, and is the estimator introduced for these chains [25]. In practice we take the lowest excitation at the smallest non-zero $|m|$ returned by the diagonalisation and divide by $|m|$; in every case examined this state has $|m| = 1$.

The parity $P = \prod_i \sigma_i^x$ also commutes with the Hamiltonian, and the energy difference in Eq. (D.4) mixes two branches unless the lowest states at $m = 0$ and $m = 1$ share a parity. We verified directly that they do, at $N = 10$ and $N = 12$ and for every coupling up to $p = 0.5$, so no parity projection is needed.

As a control before the extrapolation, we validate the conversion from a spectral gap to a scaling dimension using the anisotropic XY chain,

$$H_{\mathrm{XY}} = -\sum_j \left[\frac{1+\gamma}{2}\,\sigma_j^x\sigma_{j+1}^x + \frac{1-\gamma}{2}\,\sigma_j^y\sigma_{j+1}^y\right] - \lambda\sum_j \sigma_j^z, \tag{D.5}$$

written with the coupling on σ^x and the field on σ^z, the opposite convention to Eq. (13); at $\gamma = 1$ it reduces to the transverse-field Ising chain. The transformation of Appendix C maps it to free fermions with dispersion

$$\varepsilon(k) = 2\sqrt{(\lambda - \cos k)^2 + \gamma^2\sin^2 k}. \tag{D.6}$$

At $\lambda = 1$ the gap closes at $k = 0$, which is what puts the chain at criticality. Expanding there with $1 - \cos k \simeq k^2/2$ and $\sin k \simeq k$ gives $\varepsilon \simeq 2|k|\sqrt{\gamma^2 + k^2/4}$, so $\varepsilon \to 2\gamma|k|$ as $k \to 0$ and the velocity is exactly $v(\gamma) = 2\gamma$. At $\gamma = 1$ this returns the Ising value $v = 2$ used throughout. The leading boundary dimension remains $x_1 = \frac{1}{2}$. The test runs through Eq. (7), which gives the first gap as $\mathrm{Im}(\lambda_1 - \lambda_0) = \pi x_1/(vT)$. Its leading behaviour is thus $1/T$; the terms linear in T and constant in T are common to both eigenvalues and cancel in the difference, so the next correction is $1/T^3$. We therefore fit

$$\mathrm{Im}(\lambda_1 - \lambda_0) = \frac{a_1}{T} + \frac{a_3}{T^3}, \tag{D.7}$$

whose leading coefficient is $a_1 = \pi x_1/v$. Both quantities on the right are known: the initial product state realises the same free boundary as in Section 2.1, so $x_1 = \frac{1}{2}$, and $v = 2\gamma$ from the expansion of Eq. (D.6), which gives the exact value $a_1 = \pi/(4\gamma)$. Read the other way round, $x_1 = v\,a_1/\pi$ turns a measured slope into a dimension. As shown in Table 2, using the correct velocity gives $x_1 = 0.461$, 0.497, and 0.496. Including the coupling-dependent velocity therefore keeps the extracted dimension close to $\frac{1}{2}$.

Returning to the ANNNI-type chain, the accessible ring sizes are far from the thermodynamic limit: at $p = 0$ the $N = 18$ estimate is 1.9899 against the exact value 2, and

γ	$v = 2\gamma$	a_1	$\pi/4\gamma$	x_1 from $v = 2\gamma$
0.5	1	1.449	1.571	0.461
1.0	2	0.781	0.785	0.497
1.5	3	0.520	0.524	0.496

Table 2: **Validation on the anisotropic XY chain.** The first spectral gap is fitted as $\mathrm{Im}(\lambda_1 - \lambda_0) = a_1/T + a_3/T^3$ over the converged times: $T = 2, 3, 4$ for $\gamma = 0.5$ and 1.0, and $T = 2, 4$ for $\gamma = 1.5$. The exact coefficient is $a_1 = \pi/(4\gamma)$ for $x_1 = \frac{1}{2}$. The final column converts the fitted coefficient using the correct velocity $v = 2\gamma$.

reaching three-digit accuracy directly would need about $N = 57$. We therefore extrapolate, and fix the form of the correction at the integrable point, where the answer is known. Writing the error as $2 - v(N) \propto N^{-\alpha}$, the rescaled quantity $[2 - v(N)]\, N^{\alpha}$ is constant for $\alpha = 2$ alone, drifting downwards for $\alpha = 1$ and upwards for $\alpha = 3$.

We therefore fit

$$v(N) = v_\infty + \frac{C}{N^2} \tag{D.8}$$

at every coupling and use v_∞ throughout; Table 3 lists the data and fit parameters, and Figure 18 shows the extrapolations. At $p = 0$ the extrapolated value lies within 6×10^{-5} of the exact result. As the coupling grows, this extrapolation is more important since $|C|$ rises from 3.27 at $p = 0$ to 140.7 at $p = 1.5$.

p	$N = 10$	$N = 12$	$N = 14$	$N = 16$	$N = 18$	v_∞	C
0.0	1.9673	1.9772	1.9833	1.9872	1.9899	1.99994	−3.27
0.1	2.5707	2.6007	2.6191	2.6312	2.6396	2.67016	−9.97
0.2	3.1565	3.2073	3.2387	3.2594	3.2738	3.32561	−16.96
0.3	3.7247	3.7972	3.8423	3.8722	3.8931	3.96723	−24.34
0.4	4.2756	4.3707	4.4305	4.4703	4.4981	4.59585	−32.16
0.5	4.8096	4.9284	5.0038	5.0543	5.0897	5.21214	−40.46
1.0	7.2534	7.5020	7.6674	7.7824	7.8653	8.12650	−88.23
1.5	9.4140	9.7946	10.0594	10.2504	10.3921	10.79924	−140.69

Table 3: **Finite-size velocities and thermodynamic extrapolations.** Values of $v(p)$ at $\lambda = 1$ for periodic chains of $N =$ 10–18, with the parameters of the fit $v(N) = v_\infty + C/N^2$. The conformal analysis uses the extrapolated values v_∞.

Over the range studied the extrapolated velocity is close to linear in p, $v_\infty \simeq 2.02 + 6.42p$, which is the trend shown in Figure 5. The linearity is only approximate: the slope between successive couplings falls steadily from 6.70 near $p = 0$ to 5.35 between $p = 1.0$ and $p = 1.5$, so the growth flattens at stronger coupling, and a fit at fixed system size still drifts with N. We therefore treat the relation as a description of the trend, and every conversion in this thesis uses the separately extrapolated $v_\infty(p)$.

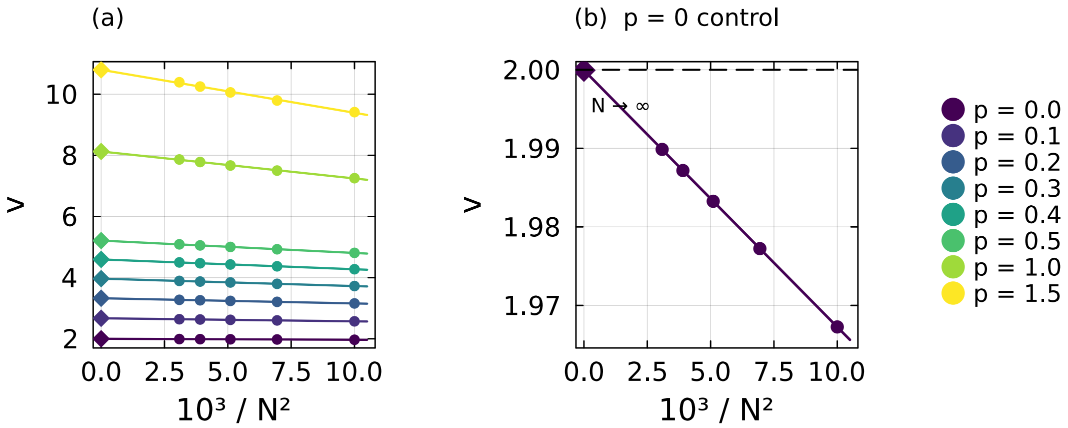


Figure 18: **Extrapolation of the sound velocity.** (a) Finite-size estimates of v against $1/N^2$ at each coupling, with fitted lines and thermodynamic intercepts (diamonds). (b) The integrable point on an expanded scale. The estimates are linear in $1/N^2$, and the intercept $v_\infty = 1.99994$ agrees with the exact value $v = 2$ (dashed line).

E Construction of the time-evolution MPO

This appendix describes the time-evolution MPO used in Section 5. We give its construction from the Hamiltonian MPO, the explicit local tensor of the second-order scheme, and numerical comparisons with the lower-order alternatives.

The construction begins from the Hamiltonian represented as an MPO. We supply the Hamiltonian as a symbolic sum of operator strings, from which the library assembles the FSM of Section 5.1. At each bond, it collects the terms that cross the bond, arranges their coefficients into a matrix, and compresses this matrix with an SVD. The retained components form the minimal set of memory channels required to carry the interactions across the bond. We denote their number by χ_H to distinguish it from the bond dimension χ used to truncate the temporal states. For the ANNNI-type Hamiltonian, the compression gives $\chi_H = 3$, corresponding to the three intermediate states in Eq. (15), and hence $D_W = \chi_H + 2 = 5$. It also determines the four blocks in that equation: D contains the on-site term, C initiates an interaction, B terminates it, and A propagates it across a site.

We construct the local evolution tensor with `ITensorExpMPO.jl` [58], which is built on `ITensors.jl` [59] and implements the operators W^{I} and W^{II} of Zaletel et al. [51]. Under the single-layer application considered here, neither operator is second-order accurate, as shown below. We therefore added the second-order cluster construction of Van Damme et al. [50]. The implementation is available in the thesis repository [60], together with its documentation.

E.1 The second-order *VD2* construction

We seek a local tensor $W(\tau)$, with $\tau = -i\delta t$, whose product along the chain approximates $e^{\tau H}$. For a Hamiltonian $H = \sum_x H_x$, the expansion is

$$e^{\tau H} = \mathbb{1} + \tau \sum_x H_x + \frac{\tau^2}{2} \sum_{x,y} H_x H_y + \dots \tag{E.1}$$

Products with disjoint support commute and factorise, so a repeated local tensor generates them without additional memory channels. Products of overlapping terms, on the other hand, require a more explicit treatment. Because the corresponding operators act on a common site, they need not commute, and both H_xH_y and H_yH_x must appear with the weights prescribed by Eq. (E.1). The order of the evolution MPO is therefore determined by the overlapping products retained in its local tensor.

The constructions W^{I} and W^{II} carry at most one interaction across each bond and have virtual bond dimension $1+\chi_H=4$ for the present Hamiltonian [51]. The operator W^{I} retains only products with disjoint support. The operator W^{II} additionally includes products that overlap on one site and reproduces the on-site field to all orders. Neither construction retains products in which two interactions cross the same bond. Since these terms first appear at order δt^2, a single application of either operator is only first-order accurate for the NNN model we study in this work. Appendix E.2 verifies this scaling numerically.

Second-order accuracy can be obtained either by composing several stages [51] or by retaining the missing overlapping pairs in one application. A staged composition is efficient in conventional time evolution because its sub-steps together advance a single interval δt. In the transverse contraction, however, all stages must be blocked into one repeated column. After rotation, two stages produce a local physical dimension $4^2=16$, compared with 13 for the single-stage second-order construction used here.

The three constructions can be understood as modifications of the FSM in Section 5.1. Rather than terminating at the completed level, the machine returns to its starting level with a factor τ, after which the completed level is removed. What is removed is the level itself, not the transitions that reached it: those now reach the starting level instead, so each completed interaction along a walk contributes its own factor of τ. The resulting operator is a sum over walks through the machine, with one transition applied at each lattice site. A valid walk begins and ends at the first level, may remain there through the identity loop, and acquires one factor of τ for each completed interaction. Expanding the local tensor therefore produces the identity, the Hamiltonian, and all disjoint products with the coefficients of Eq. (E.1). Overlapping pairs remain absent because the machine carries only one interaction at a time.

The missing pairs are recovered by applying the same modification to the machine representing H^2, whose levels are pairs of levels from the original machine. Figure 19 illustrates this construction. The levels $(1,3)$ and $(3,1)$ describe configurations in which one term is complete and the other has not started. They contain disjoint products and fold back to the initial level with a factor $\tau/2$. After these sectors are removed, the level $(3,3)$ can be reached only by overlapping terms. Folding it back with $\tau^2/2$ supplies the connected pairs omitted by W^{I} and W^{II}. The general construction at arbitrary order, including the exact compressions used below, is given by Van Damme et al. [50].

Table 4 lists the transitions of the pair machine in terms of the blocks in Eq. (15), while Figure 19(b) shows the corresponding level structure and folds. At each site, either one index advances, both indices advance through a composite transition, or the pair remains on a self-loop.

The overlapping interactions $\sigma_3^z\sigma_4^z$ and $\sigma_4^z\sigma_5^z$ provide a concrete example of the difference between the two machines. In the single-interaction machine of Figure 19(a), site 3 applies C to open the first term. At site 4, the walk would then need to apply both B, which closes the first term, and C, which opens the second. No such transition exists because each site applies one move and the machine carries only one interaction. After returning to the first level, the earliest site at which the second term can open is site 5. Products generated by successive completions are therefore disjoint, with one factor of τ for each completed term. The overlapping product is absent at order τ^2, producing the

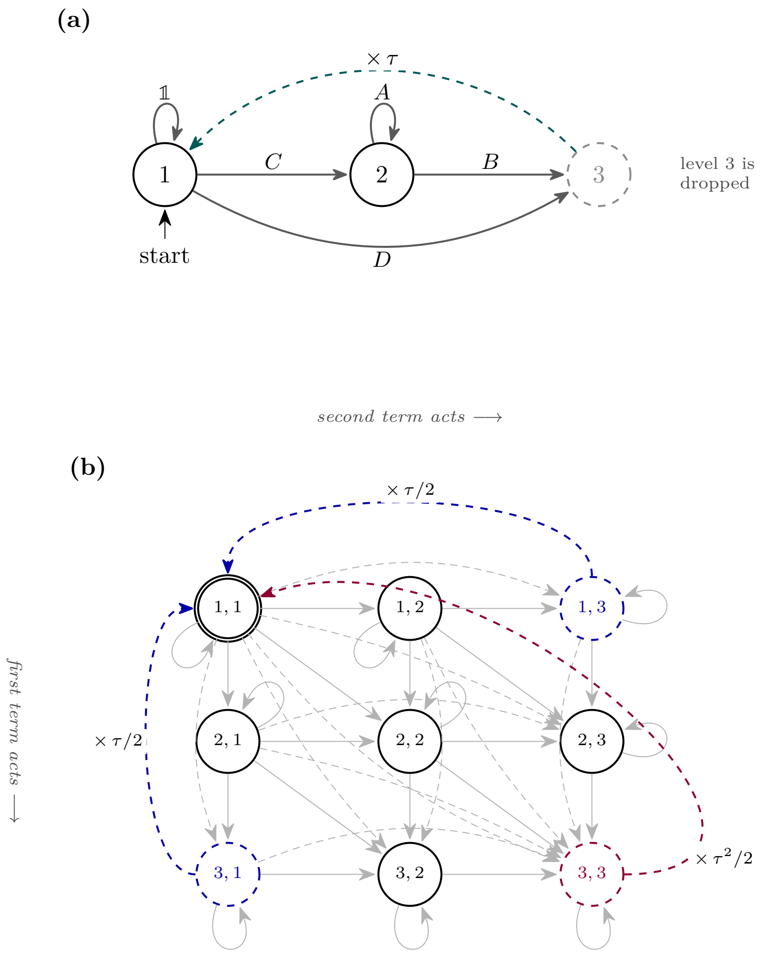


Figure 19: **Construction of the evolution operator from the Hamiltonian machine.** (a) The machine of Figure 6, reduced to three levels. Returning to level 1 with a factor τ (teal), rather than terminating at level 3, gives a first-order approximation to $e^{\tau H}$; the completed level is then removed (grey). (b) The pair machine for H^2, with one level index for each Hamiltonian term. Grey arrows advance either one index or both indices at the same site. Each level also has a self-loop formed from the identity at level 1, the memory operator A at level 2, or their pair products. Densely dashed arrows denote the on-site term, which advances an index directly to level 3. The disjoint sectors $(1,3)$ and $(3,1)$ return to the initial level with $\tau/2$ (blue). The remaining sector $(3,3)$ contains the overlapping pairs and returns with $\tau^2/2$ (purple).

leading error of W^{I} and W^{II}.

The pair machine in Figure 19(b) carries both interactions simultaneously. The corresponding walk uses $(1,1) \to (2,1)$ through C at site 3, followed by $(2,1) \to (3,2)$ through the composite entry BC at site 4. This transition closes the first interaction while opening the second at the shared site. The move $(3,2) \to (3,3)$ through B at site 5 then completes the pair. Folding $(3,3)$ back to the initial level supplies the factor $\tau^2/2$, while exchanging the two copies generates the opposite operator ordering with the same weight. Thus, disjoint products acquire one factor of τ per completion, whereas each overlapping pair receives the required second-order weight from a single fold.

Retaining clusters of up to n overlapping terms gives a one-step error of order δt^{n+1}. The first-order member is W^{II}, while the second-order construction used here reproduces every connected pair. Representing two interactions that cross the same bond requires

from \ to	$(1,1)$	$(1,2)$	$(1,3)$	$(2,1)$	$(2,2)$	$(2,3)$	$(3,1)$	$(3,2)$	$(3,3)$
$(1,1)$	$\mathbb{1}$	C	D	C	CC	CD	D	DC	DD
$(1,2)$		A	B		CA	CB		DA	DB
$(1,3)$			$\mathbb{1}$			C			D
$(2,1)$				A	AC	AD	B	BC	BD
$(2,2)$					AA	AB		BA	BB
$(2,3)$						A			B
$(3,1)$							$\mathbb{1}$	C	D
$(3,2)$								A	B
$(3,3)$									$\mathbb{1}$

Table 4: **Transitions of the H^2 pair machine.** Rows specify the current pair of levels and columns the next pair. The row and the column fix one move for each copy, and the entry is the product of the two corresponding blocks of Eq. (15) applied at a single site, with the first copy written first. The blocks multiply as operators on the physical index and form a direct product on the memory indices. Empty entries denote forbidden transitions, while diagonal entries are the self-loops shown in Figure 19(b). Folding adds the $(1,3)$ and $(3,1)$ columns to the $(1,1)$ column with weight $\tau/2$, and the $(3,3)$ column with weight $\tau^2/2$, before the corresponding rows and columns are removed.

a doubled-memory sector of dimension χ_H^2, in addition to the scalar and single-memory sectors. The resulting virtual bond dimension is

$$1 + \chi_H + \chi_H^2. \tag{E.2}$$

For the ANNNI-type model, this gives 13, compared with 4 for W^{I} and W^{II}. At $p = 0$, only the NN interaction remains and $\chi_H = 1$, giving bond dimensions 3 and 2, respectively.

Before combining the blocks, we distribute the time-step factor asymmetrically between the interaction endpoints. The initiating block carries $t_C = \sqrt{\delta t}$, while the terminating block carries $t_B = \tau/t_C$, so every completed interaction acquires one factor of τ:

$$\tilde{D} = \tau D, \qquad \tilde{C} = t_C\, C, \qquad \tilde{B} = t_B\, B, \qquad \tilde{A} = A. \tag{E.3}$$

This division amounts to a rescaling of the virtual index. The two endpoint factors combine to give the required factor τ for every completed interaction, leaving the physical matrix elements unchanged while keeping the tensor entries balanced at small δt.

We express the local representation of $e^{\tau H}$ through symmetrised products of the scaled blocks. The notation $\{X_1 \dots X_m\}$ denotes the sum over their distinct orderings; for example, $\{XY\} = X \otimes Y + Y \otimes X$, $\{XX\} = X \otimes X$, and $\{XXY\} = XXY + XYX + YXX$. Here $\otimes$ is the Kronecker product on the virtual memory index, combined with ordinary operator multiplication on the physical index. The numerical prefactors are the factorial weights of the exponential expansion. The expressions below specialise the second-order cluster operator of Van Damme et al. [50] to the blocks in Eq. (15).

The local tensor has a 3×3 block structure, distinct from the grid in Figure 19(b). In that grid, the two indices label the levels of the two Hamiltonian copies. Here the indices 1, 2, and 3 label the scalar, single-memory, and doubled-memory sectors obtained after equivalent levels are merged:

$$W^{VD2} = \begin{pmatrix} W_{11} & W_{12} & W_{13} \\ W_{21} & W_{22} & W_{23} \\ W_{31} & W_{32} & W_{33} \end{pmatrix}, \tag{E.4}$$

with

$$W_{11} = \mathbb{1} + \tilde{D} + \tfrac{1}{2}\{\tilde{D}\tilde{D}\} + \tfrac{1}{6}\{\tilde{D}\tilde{D}\tilde{D}\}, \tag{E.5}$$

$$W_{12} = \tilde{C} + \tfrac{1}{2}\{\tilde{C}\tilde{D}\} + \tfrac{1}{6}\{\tilde{C}\tilde{D}\tilde{D}\}, \qquad W_{13} = \{\tilde{C}\tilde{C}\} + \tfrac{1}{3}\{\tilde{C}\tilde{C}\tilde{D}\}, \tag{E.6}$$

$$W_{21} = \tilde{B} + \tfrac{1}{2}\{\tilde{B}\tilde{D}\} + \tfrac{1}{6}\{\tilde{B}\tilde{D}\tilde{D}\}, \qquad W_{31} = \tfrac{1}{2}\{\tilde{B}\tilde{B}\} + \tfrac{1}{6}\{\tilde{B}\tilde{B}\tilde{D}\}, \tag{E.7}$$

$$W_{22} = \tilde{A} + \tfrac{1}{2}(\{\tilde{B}\tilde{C}\} + \{\tilde{A}\tilde{D}\}) + \tfrac{1}{6}(\{\tilde{C}\tilde{B}\tilde{D}\} + \{\tilde{A}\tilde{D}\tilde{D}\}), \tag{E.8}$$

$$W_{23} = \{\tilde{A}\tilde{C}\} + \tfrac{1}{3}(\{\tilde{A}\tilde{C}\tilde{D}\} + \{\tilde{C}\tilde{C}\tilde{B}\}), \qquad W_{32} = \tfrac{1}{2}\{\tilde{A}\tilde{B}\} + \tfrac{1}{6}(\{\tilde{A}\tilde{B}\tilde{D}\} + \{\tilde{B}\tilde{B}\tilde{C}\}), \tag{E.9}$$

$$W_{33} = \{\tilde{A}\tilde{A}\} + \tfrac{1}{3}(\{\tilde{A}\tilde{B}\tilde{C}\} + \{\tilde{A}\tilde{A}\tilde{D}\}). \tag{E.10}$$

The first row and column open and close the memory sectors, while W_{11} contains the on-site evolution through third order. Nine of the thirteen virtual states belong to the doubled-memory sector. This sector allows two interactions to cross the same bond simultaneously and therefore represents the overlapping pairs omitted by W^{II}.

E.2 Comparing the two constructions in practice

We first assess numerical stability by measuring norm preservation under repeated application of each MPO. We apply the operators to $|X{+}\rangle^{\otimes N}$ on a chain of $N = 100$ sites at $p = 0.1$, using $\delta t = 0.05$ and a truncation cutoff of 10^{-12} after each layer. Table 5 reports the resulting norms.

	W^{I}	W^{II}	*VD2*
virtual bond	4	4	13
$\lVert\psi\rVert$ after 1 layer	1.3514	1.1341	0.99995
$\lVert\psi\rVert$ after 8 layers	12.6146	3.0181	0.99923
$\lVert\psi\rVert$ after 16 layers	307.3900	10.8328	0.99751

Table 5: **Norm drift under repeated application.** Norm of the evolved state $|X{+}\rangle^{\otimes 100}$ after the stated number of layers at $p = 0.1$ and $\delta t = 0.05$, using a truncation cutoff of 10^{-12} after each layer. Exact time evolution preserves the norm, so deviations from unity measure the error accumulated by the approximate operator.

The two first-order constructions accumulate substantial norm errors. After sixteen layers, W^{I} increases the norm by more than two orders of magnitude, while W^{II} increases it by approximately one order of magnitude. By contrast, the *VD2* result remains within 2.5×10^{-3} of unity. An evolution to $T = 10$ at $\delta t = 0.05$ requires approximately two hundred layers, making the first-order constructions unsuitable at this time step.

Norm preservation measures accumulated stability but does not determine the largest time step compatible with a given accuracy. We therefore evaluate the state error at fixed physical time. For $N = 12$, the Hamiltonian can be diagonalised exactly. We evolve $|X{+}\rangle^{\otimes N}$ to $T = 1$ using $T/\delta t$ layers in the Schrödinger picture and measure the distance from the exact state at $p = 0.1$. This comparison isolates the evolution operator from the transverse-contraction procedure. Table 6 reports the errors and effective convergence orders.

A construction of order q has a one-layer error of order $\delta t^{\,q+1}$. Reaching a fixed physical time T requires $T/\delta t$ layers, so accumulation gives a total error of order $T\,\delta t^{\,q}$ [12]. Halving the time step should therefore reduce the error by 2^q, and the base-two logarithm of the ratio between successive errors estimates q. Higher-order contributions affect this estimate at larger time steps, but their influence decreases with δt.

	W^{I}		W^{II}		*VD2*	
δt	error	order	error	order	error	order
0.2	2.01×10^{1}	–	2.54	–	1.09×10^{-1}	–
0.1	4.43	2.18	9.29×10^{-1}	1.45	2.44×10^{-2}	2.16
0.05	1.36	1.71	3.93×10^{-1}	1.24	5.93×10^{-3}	2.04
0.025	5.31×10^{-1}	1.35	1.80×10^{-1}	1.12	1.47×10^{-3}	2.01
0.0125	2.36×10^{-1}	1.17	8.65×10^{-2}	1.06	3.68×10^{-4}	2.00
0.00625	1.12×10^{-1}	1.08	4.24×10^{-2}	1.03	9.24×10^{-5}	2.00

Table 6: **Convergence towards the exact state.** Distance from the exactly evolved state of an $N = 12$ chain at fixed physical time $T = 1$ and $p = 0.1$, as a function of the time step. The effective order is the base-two logarithm of the ratio between successive errors and approaches the asymptotic convergence order as δt decreases.

For *VD2*, the effective order is 2.04 at $\delta t = 0.05$ and remains 2.00 at smaller steps. The orders of W^{I} and W^{II} instead decrease from 1.71 and 1.24 to 1.08 and 1.03, respectively. These results confirm second-order convergence for *VD2* and first-order convergence for the other two constructions.

The larger virtual bond dimension of *VD2* is offset by the larger time step permitted at fixed accuracy. A first-order construction requires a smaller δt, increasing the temporal-chain length in inverse proportion and introducing an additional truncation at every new site. At fixed β_0 and $T = 10$, the temporal chain contains 104 sites for $\delta t = 0.1$ and 208 sites for $\delta t = 0.05$. We therefore use *VD2* with $\delta t = 0.1$ in the production calculations.

F The power-method solvers: implementation and reliability

Section 6.2 introduces the block power method, which propagates k left and k right temporal states and projects the transfer-matrix eigenproblem onto their span. In the production calculations, we use this method to resolve the subleading eigenvalues required for the boundary operator content. The dominant pair and the temporal entropies are instead obtained with the less expensive single-vector iteration. This appendix describes the implementation of the block method, the numerical settings of both solvers, the validation of the implementation against exact diagonalisation, and the reliability analysis underlying the entropy data. The routine `block_transfer_eigs` is built on `ITransverse.jl` [36]. The source code and analysis notebooks are available in the thesis repository [60].

F.1 The iteration as implemented

Each iteration updates the blocks $\{|R_j\rangle\}$ and $\{\langle L_j|\}$ through four operations: application, projection, solution of the reduced problem, and separation of the resulting states.

We first apply the transfer matrix to each right state and its transpose to each left state, requiring $2k$ operator–state products. Each application increases the temporal bond dimension, so we immediately truncate the propagated state back to the bond cap of the iteration, using either of the two schemes of Section 3.4. We then construct the overlap and projected-transfer matrices of Eq. (19),

$$S_{ij} = \langle L_i | R_j \rangle \,, \qquad M_{ij} = \langle L_i | \, \mathcal{E} \, | R_j \rangle \,. \tag{F.1}$$

Both matrices use the same bilinear contraction without complex conjugation because the left states are already represented as bras. Figure 20 shows these contractions, and Algorithm 1 summarises one iteration.

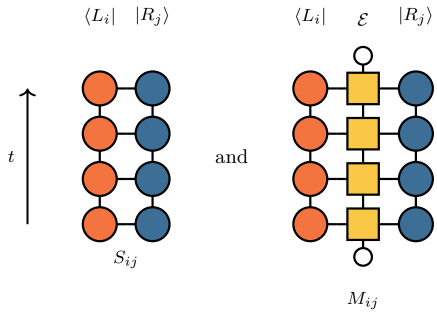


Figure 20: **Projected matrices.** Contractions defining the $k \times k$ matrices in Eq. (19). Each tMPS forms a column, while $\mathcal{E}$ is the corresponding column of the space–time network closed at both ends by the product state (small open circles). The left state enters directly as a bra, without complex conjugation.

Algorithm 1 Block power method for the k dominant eigenpairs of $\mathcal{E}$.

$|R_j\rangle$, $\langle L_j| \leftarrow$ random complex tMPSs, $\quad j = 1, \dots, k$
repeat
  $\left|\tilde{R}_j\right\rangle \leftarrow \text{truncate}_\chi\left(\mathcal{E}\,|R_j\rangle\right), \qquad \left\langle\tilde{L}_j\right| \leftarrow \text{truncate}_\chi\left(\langle L_j|\,\mathcal{E}\right)$ ▷ apply
  $S_{ij} \leftarrow \langle L_i|R_j\rangle, \qquad M_{ij} \leftarrow \left\langle L_i\middle|\tilde{R}_j\right\rangle = \langle L_i|\,\mathcal{E}\,|R_j\rangle$ ▷ project
  solve $Mv = \theta\, Sv$ for Ritz values $\theta_1, \dots, \theta_k$ and coefficients V, U
  $|R_j\rangle \leftarrow \text{truncate}_\chi\left(\sum_i V_{ij}\left|\tilde{R}_i\right\rangle\right), \qquad \langle L_j| \leftarrow \text{truncate}_\chi\left(\sum_i U_{ij}\left\langle\tilde{L}_i\right|\right)$ ▷ separate
  normalise every block member
until the leading Ritz values stop changing within tolerance

The reduced problem $Mv = \theta Sv$ becomes ill-conditioned when the block states approach linear dependence. Direct inversion of S would then amplify its smallest singular values, which are dominated by numerical noise. We instead form the Moore–Penrose pseudoinverse S^+ and retain only singular values larger than 10^{-12} times the largest. The discarded directions are mapped to zero, giving

$$W = S^+ M, \qquad W\,v_j = \theta_j\,v_j. \tag{F.2}$$

The eigenvalues θ_j, ordered by decreasing modulus, approximate $\mu_0, \dots, \mu_{k-1}$. Writing the right coefficient vectors as the columns of V and the Ritz values as the diagonal matrix Θ gives $W = V\Theta V^{-1}$.

The left and right coefficients must correspond to the same Ritz values. Solving a separate transposed problem would produce two independently ordered spectra, whose matching becomes ambiguous within a near-degenerate cluster. We therefore derive the left coefficients from the decomposition of W. After discarding the smallest singular values, S^+S and SS^+ act as the identity on the retained right and left subspaces, respectively. For $u^\mathsf{T} M = \theta u^\mathsf{T} S$, define $y^\mathsf{T} = u^\mathsf{T} S$. Since $u^\mathsf{T} S S^+ = u^\mathsf{T}$,

$$u^\mathsf{T} M = y^\mathsf{T} S^+ M = y^\mathsf{T} W. \tag{F.3}$$

Thus y^T is a left eigenvector of W. Since the left eigenvectors are the rows of V^{-1}, the coefficients are

$$u_j = \left(S^+\right)^\mathsf{T} \left(V^{-1}\right)^\mathsf{T} e_j, \tag{F.4}$$

where e_j is the j-th unit vector. The common index j fixes the left–right pairing and gives biorthogonality in the S metric, $u_i^\mathsf{T} S v_j = \delta_{ij}$.

Finally, we reconstruct the states from these coefficients,

$$|R_j\rangle \leftarrow \text{truncate}_\chi\Big(\sum_i V_{ij}\,\big|\tilde{R}_i\big\rangle\Big), \qquad \langle L_j| \leftarrow \text{truncate}_\chi\Big(\sum_i U_{ij}\,\big\langle\tilde{L}_i\big|\Big), \tag{F.5}$$

where $\big|\tilde{R}_i\big\rangle$ and $\big\langle\tilde{L}_i\big|$ are the propagated block states. We sum the tensors exactly and compress the result afterwards. Density-matrix addition of the MPSs is less expensive but becomes unstable when the states are nearly parallel, which is also where eigenvector separation is most difficult. Each reconstructed state is renormalised. If its norm underflows or overflows, we replace it with a new random vector rather than propagate the lost direction as numerical noise.

F.2 The eigenvalue-only mode

Separating the invariant subspace into individual eigenvectors is the least stable part of the iteration. As two eigenvalues approach, the coefficient matrix V becomes ill-conditioned and amplifies truncation noise in the reconstructed states. The Ritz values are less sensitive because they depend on the projected operator over the complete subspace rather than on a particular basis within it.

All spectra obtained with the block method therefore use a mode that avoids this separation. We propagate orthonormal bases of the same invariant subspaces rather than individual eigenvectors, which is sufficient to determine the Ritz values and their phase gaps. The two modes share all preceding operations and differ only in the basis update, so their comparison isolates the effect of eigenvector separation. We choose the eigenvalue-only mode for stability rather than speed: for identical seeds at $p = 0.1$, the runtimes of the two modes agree within a few per cent. After ordering the Ritz values by modulus, we replace the coefficient matrices by the orthonormal factors from their QR decompositions,

$$V \to Q_V, \qquad U \to Q_U, \qquad Q^\dagger Q = \mathbb{1}. \tag{F.6}$$

The first j columns of each Q span the same subspace as the corresponding first j coefficient vectors. The ordering of the leading subspaces is therefore retained without assigning an individual eigenvector to each block member.

This basis change leaves the Ritz values invariant. Consider invertible basis transformations $|R_j\rangle \to \sum_i G_{ij}\,|R_i\rangle$ and $\langle L_i| \to \sum_m H_{mi}\,\langle L_m|$. They transform the projected matrices as

$$S \;\to\; H^\mathsf{T} S G, \qquad M \;\to\; H^\mathsf{T} M G. \tag{F.7}$$

Because both matrices acquire the same factors, the generalised eigenproblem retains the same Ritz values. Moreover, Q_V and Q_U are unitary, so the states passed to the next truncation are not amplified by an ill-conditioned eigenvector transformation.

We compare the two modes using identical random initial states at $p = 0.1$ over $T = 2$–8. Their leading moduli and first gaps agree. Visible differences occur only among the final members of the block, whose Ritz values are furthest from the dominant value and converge most slowly. We do not use these levels as precision estimates.

F.3 Numerical parameters and controls

The following settings affect the numerical trajectory and attainable accuracy, but not the target fixed point. Unless stated otherwise, the tests use the production parameters of Section 7.

Temporal bond dimension. The converged temporal states have modest bond dimensions. At the ends of the production windows, the four-state blocks reach $\chi = 15$ at $p = 0$, $T = 14$ and $\chi = 12$ at $p = 0.1$, $T = 9$. The larger eight-state tower blocks reach $\chi = 33$, below the cap of 48 used for those runs. We nevertheless impose a cap to prevent transient growth from exhausting the available memory. At $p = 0.1$ and $T = 9$, for example, intermediate single-vector states reach the production cap of 64, while the converged pair has $\chi = 12$. Repeating these runs with a cap of 128 from the same initial states gives the same converged pair. The cap therefore affects the numerical trajectory but not the fixed point in the tested regime.

Singular-value cutoff. The block iteration uses a cutoff of 10^{-8} for its first forty iterations and 10^{-10} thereafter, while the single-vector runs use 10^{-12} throughout. Comparisons across cutoffs of 10^{-8}, 10^{-10}, and 10^{-12} give consistent eigenvalues for independent initial states. The cutoff therefore affects the stopping condition more strongly than the result. A looser cutoff keeps the states smaller and reduces the cost per iteration, but raises the truncation noise floor and causes more runs to end through the no-improvement rule. A tighter cutoff retains more singular values and is more expensive, but reaches the formal tolerance more often.

Stopping rule. We assess convergence from the two leading Ritz values, matching each to its nearest predecessor in the complex plane. Under emergent dual unitarity, the leading moduli approach one another. At $p = 0$ and $T = 20$, the three largest moduli barely differ, so truncation noise can change their ordering even after the individual values have stabilised. The iteration stops when the matched change falls below 10^{-6} or when no further improvement occurs within 150 iterations. With RTM truncation, the change can reach a noise floor above the formal tolerance; the second condition terminates these runs.

Block size. The required block size is determined by the number of states rather than by the number of distinct scaling dimensions because conformal descendants produce degeneracies within a tower. The fixed-boundary Ising tower begins $2, 3, 4, 4, 5, 5, \ldots$ (Appendix G.2). A four-state block cuts through the degenerate pair at 4. Independent sequences then disagree on the first gap at approximately half of the evolution times, despite agreeing on the leading eigenvalue. With $k = 8$, the retained subspace ends at a genuine spectral gap and this disagreement disappears. We therefore use $k = 8$ for the boundary-spectrum calculations and $k = 4$ for the leading-eigenvalue sequences in Table 1.

Trotter step. The second-order evolution operator keeps the discretisation error small at the production value $\delta t = 0.1$. Halving the time step while doubling N_β, and hence keeping β_0 fixed, leaves the entropy plateau unchanged within the reliable window. It also does not change the physical time at which runs begin to be rejected. The smaller step doubles the temporal-chain length at fixed physical time, increasing both the computational cost and the number of truncations in each sweep.

Cost. Two operations dominate each block iteration: the $2k$ operator–state products required to apply the transfer matrix and the $2k$ linear combinations of k states used to rebuild the blocks. Before compression, each combination carries the sum of the bond dimensions of its terms. The single-vector iteration avoids these combinations and is therefore substantially less expensive. We use it for the dominant pair and all temporal entropies. In production, each block is warm-started from the converged result at the preceding half-unit time increment.

F.4 Validation against exact diagonalisation

The block method introduces two approximations: it retains only k directions of the transfer matrix, and it truncates every propagated state to finite bond dimension. Neither approximation provides a directly accessible error bound. We therefore validate the implementation at short evolution times, where the rotated transfer matrix can be constructed as a dense array and diagonalised without state truncation or subspace projection.

Table 7 compares the two leading eigenvalues in three cases with the same numerical settings but different couplings and numbers of temporal sites. Without cooling sites, each temporal site represents one Trotter step. At $\delta t = 0.5$, three sites therefore correspond to $T = 1.5$, while four sites correspond to $T = 2$. The local physical dimension equals the bond dimension of the evolution MPO: 13 at $p = 0.1$ and 3 at $p = 0$. A dense transfer operator on n temporal sites consequently has dimension d^n. This exponential growth restricts the validation to short times.

The cases differ mainly in their stopping condition. The 27-dimensional calculation reaches the required tolerance, and its two leading moduli agree with the dense result to 2.5×10^{-12}, showing that the reduced eigenproblem introduces no measurable error. The two larger calculations reach the noise floor described in Section F.3 and terminate through the no-improvement rule. Their leading moduli agree with the dense results to within 10^{-7} for dimensions 81 and 2197. The attainable accuracy is therefore controlled by convergence of the iteration rather than by the projection onto the block.

This comparison validates the complete short-time pipeline: construction of the evolution operator, extraction of the rotated column, projection onto the block, and solution of the reduced problem. It does not validate the late-time regime, where the leading eigenvalues become nearly degenerate and the dense matrix is no longer computationally accessible.

operator	dim.	exact $\lvert\mu_0\rvert$, $\lvert\mu_1\rvert$	error	stopped on
critical Ising, $T = 1.5$	27	$0.9048,\ 0.8373$	2.5×10^{-12}	tolerance
critical Ising, $T = 2$	81	$0.9129,\ 0.8702$	2.3×10^{-7}	no improvement
ANNNI-type $p = 0.1$, $T = 1.5$	2197	$0.9405,\ 0.9082$	1.2×10^{-7}	no improvement

Table 7: **Validation against exact diagonalisation.** Comparison of the block method with dense diagonalisation of the same rotated transfer operator at $\delta t = 0.5$, without cooling sites. The error is the largest absolute deviation in the moduli of the two leading Ritz values. None of the three cases reaches the bond cap. The final column records the stopping condition for each run.

F.5 The limits of the entropy route

The entropy and spectral routes require different information from the transfer matrix. The spectral analysis uses the Ritz values of the leading invariant subspace, whereas the entropy requires the corresponding individual left and right eigenvectors. Eigenvalues obtained by subspace projection are second-order accurate in the error of the spanning vectors. Individual eigenvectors, however, become increasingly sensitive to the separation between neighbouring eigenvalues [61]. Emergent dual unitarity reduces these separations as the evolution time grows, causing the entropy calculation to lose reliability earlier.

We quantify this loss using independent random initialisations of the single-vector iteration. At each accessible time, we perform up to twenty runs with the production settings above; the ensembles are smaller at the longest times because each run is considerably more expensive. For the present analysis, a run is rejected when its imaginary plateau falls outside the band $(0.05, 0.20)$ around $\pi c/16 = 0.098$. This band is used only to count rejections and does not determine which times enter the fits in Table 1; those are selected by the criteria described in Section G.1. Table 8 reports the rejection counts, the median plateau of the surviving runs, and their span, defined as the ratio of the largest surviving plateau to the smallest.

p	T	rejected	plateau of surviving runs	span
0	2–7	0/20	0.1284–0.1274	1.0
0	8–17	0/20	0.1252–0.1137	1.0–1.1
0	18	1/20	0.1144	1.2
0	20	0/20	0.1146	1.8
0	22	2/20	0.1080	1.7
0.1	4–10	0/10	0.1322–0.1204	1.0–1.2
0.1	11	2/10	0.1199	1.2
0.1	12	0/10	0.1172	1.7
0.1	13	0/10	0.1158	1.5
0.1	14	1/6	0.1063	2.1
0.1	15	0/6	0.1485	2.4
0.1	16	1/6	0.1337	2.4
0.1	17	0/4	0.0699	1.4
0.3	2–6	0/10	0.1422–0.1264	1.0
0.3	7	2/10	0.1270	1.1
0.3	8	6/20	0.1212	3.4
0.3	9	11/20	0.1183	2.3
0.3	10	6/10	0.1007	2.5
0.3	11	8/10	0.1137	1.2
0.3	12	8/10	0.1569	1.7
0.5	2–3	0/10	0.1449–0.1354	1.0
0.5	3.5	2/10	0.1352	1.0
0.5	4	3/10	0.1351	1.1
0.5	4.5	6/10	0.1341	1.0
0.5	5	7/10	0.1310	1.1
0.5	6	10/10	–	–
0.5	7	2/10	0.0530	3.0
0.5	8	1/8	0.0557	2.4

Table 8: **Rejection rate of the single-vector iteration.** Independent random initialisations at each evolution time, with the ensemble size as the denominator. A run is rejected when its imaginary plateau falls outside $(0.05, 0.20)$. The last two columns give the median plateau of the surviving runs and their span. Where more than half the ensemble is rejected, too few runs remain to estimate the plateau.

Rejections occur at the level of individual runs. At a fixed time, some runs return an unphysical plateau while others remain consistent with neighbouring times. The rejection rate increases with T at every coupling, and the first rejection moves from $T = 18$ at $p = 0$ to $T = 11$, $T = 7$, and $T = 3.5$ as p increases.

The rejected runs exhibit two distinct behaviours. In the first, the ensemble scatters broadly. At $p = 0.3$ for $T \geq 11$, the returned plateaux can differ in sign, and the few values inside the band do not constitute a reliable measurement. In the second, the ensemble agrees on an unphysical value. At $p = 0.5$ and $T = 7$, six of ten runs agree within 3% but return approximately half of $\pi c/16$; the runs at $T = 8$ return the same branch.

Agreement across seeds is therefore insufficient on its own. Continuity in time reveals the failure because the surviving plateau decreases from 0.131 at $T = 5$ to 0.053 at $T = 7$.

The scatter increases before an ensemble becomes unusable. At $p = 0.3$, the surviving plateaux span 10% at $T = 7$ but a factor of three at the next time point. An ensemble is therefore required both to reject unphysical runs and to estimate the variability of the remaining values.

These ensembles also explain why the spectral route reaches longer evolution times than the entropy route. Even where no runs are rejected, the leading eigenvalues agree within 0.1% across different seeds, whereas the entropy plateaux can vary by up to 30%. Across all tested cases, the spread in eigenvalues is between ten and a thousand times smaller than that of the plateaux. The eigenvalue-only block iteration exhibits similarly weak seed dependence: six independent initialisations at $p = 0.3$ and $T = 6$–7 agree in both modulus and phase to within 0.01%. Consequently, the spectral route remains reliable well after the entropy route breaks down, accounting for the wider time windows in Table 1.

The stopping condition does not distinguish physical from unphysical runs. The single-vector iteration monitors $\mathrm{d}s$, the change in the singular values of the temporal state between successive iterations, and compares it with the convergence tolerance. With RTM truncation, $\mathrm{d}s$ initially decreases and then reaches a floor above the tolerance at nearly every time beyond $T \approx 10$, for accepted and rejected runs alike. Since each truncation introduces an error set by its cutoff, reaching this floor does not by itself imply that the result is unreliable.

Figure 21 illustrates this behaviour at one time point. With RTM truncation, $\mathrm{d}s$ decreases by a factor of 10^3 and then remains above the tolerance until the no-improvement rule terminates the run. Repeating the calculation from the same random state with RDM truncation allows $\mathrm{d}s$ to cross the tolerance and the iteration to terminate normally. The two runs return leading eigenvalues that agree within 0.01%. The noise floor therefore originates from the truncation scheme rather than from the transfer matrix, and termination by the no-improvement rule is not by itself a reason to reject a run.

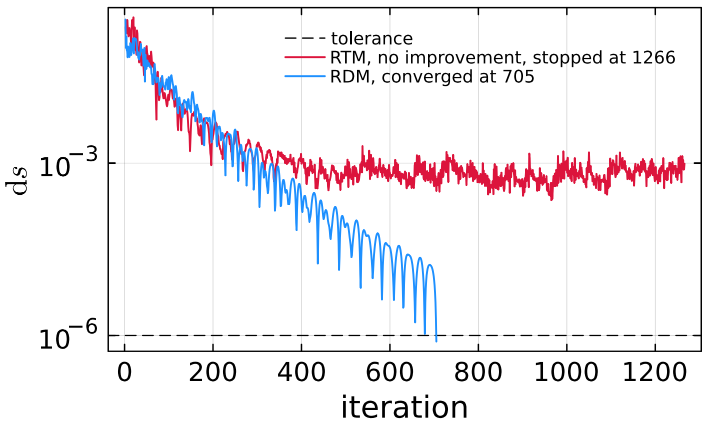


Figure 21: **Termination of the single-vector iteration.** The change $\mathrm{d}s$ as a function of iteration number at $p = 0$ and $T = 12$, for both truncation schemes started from the same random state. With RTM truncation, $\mathrm{d}s$ reaches a floor above the tolerance and the no-improvement rule terminates the run. With RDM truncation, it crosses the tolerance. The two calculations return leading eigenvalues that agree within 0.01%.

Table 9 compares selected time points under the two truncation schemes of Section 3.4. The RDM iteration reaches the tolerance at every tested time and reproduces the RTM plateau within the observed ensemble spread. Its computational cost is approximately an order of magnitude larger. We therefore use RTM in production and retain RDM as a control.

p	T	plateau		stopping		time (s)	
		RTM	RDM	RTM	RDM	RTM	RDM
0	8	0.1252	0.1248	converged	converged	12	147
0	12	0.1191	0.1211	no improvement	converged	53	785
0	14	0.1186	0.1185	no improvement	converged	98	1466
0	17	0.1137	0.1162	no improvement	converged	177	2627
0	20	0.1146	0.1145	no improvement	converged	283	5177
0	22	0.1080	0.1134	no improvement	converged	402	5125
0.1	11	0.1199	0.1207	no improvement	converged	1316	43692

Table 9: **Comparison of the two truncation schemes.** The RTM entries are medians over the surviving ensembles in Table 8. The stopping columns identify the termination condition, and the final two columns give the median runtime.

The strong coupling dependence of the eigenvector conditioning persists without tensor-network truncation. At short times, we construct the rotated transfer operator as a dense matrix and diagonalise it exactly, following Section F.4. Table 10 compares the condition number of the complete eigenvector matrix at matched dense dimension, without cooling sites and with $\delta t = 0.5$. Between $p = 0$ and the interacting cases, the condition number increases by seven to nine orders of magnitude, while the separation between the two leading eigenvalues remains comparable. The poor conditioning at $p \neq 0$ therefore cannot be explained solely by an approaching leading degeneracy. The two columns measure different things: the separation controls how strongly a perturbation mixes that particular pair of eigenvectors, whereas $\mathrm{cond}(V)$ measures how far the complete eigenvector basis is from linear independence, which for a non-normal operator can be large even when no two eigenvalues are close. Its microscopic dependence on the interaction remains unclear.

p	T	dimension	$\lvert\mu_0 - \mu_1\rvert$	$\mathrm{cond}(V)$
0	3.5	2187	0.21	3.0×10^2
0.1	1.5	2197	0.30	2.2×10^{11}
0.3	1.5	2197	0.19	2.1×10^{10}
0.5	1.5	2197	0.14	9.6×10^9

Table 10: **Conditioning of the exact eigenvector basis.** Exact diagonalisation of the rotated transfer operator at matched dense dimension, without cooling sites and with $\delta t = 0.5$. Here $\mathrm{cond}(V)$ is the condition number of the eigenvector matrix. The gap $\lvert\mu_0 - \mu_1\rvert$ remains comparable across the couplings, whereas the conditioning changes by several orders of magnitude.

The entropy route fails when the individual eigenvectors can no longer be resolved reliably. This occurs earlier than the loss of the spectral Ritz values because eigenvectors are sensitive to the separation between neighbouring eigenvalues, which emergent dual unitarity reduces with time. The coupling controls the onset: the first rejection moves from $T = 18$ at $p = 0$ to $T = 3.5$ at $p = 0.5$, following the increase in eigenvector conditioning rather than the closing of the leading gap. Since a single run provides no reliable warning near this boundary, ensembles are required to detect failures and determine the entropy uncertainties in Table 1.

Some questions remain. First, the mechanism by which the coupling controls the eigenvector conditioning is unknown. Second, no diagnostic from a single run identifies a degraded entropy result near the failure boundary, so the spectral windows in Table 1 still require independent repetitions and the phase-increment criterion of Section 6.3.

G Supporting measurements, corrections and uncertainties

This section provides the supporting analysis for Section 7. It presents the entropy profiles at each coupling, the finite-time corrections and uncertainties in both routes, and the transfer-matrix spectra for different pairs of boundary conditions.

G.1 Corrections and uncertainties

Neither route determines the central charge directly. The measured quantities approach their universal values only as $T \to \infty$, so both estimates require a finite-time correction and an extrapolation. Their reliability is controlled primarily by the length of the available time window, rather than by the precision of an individual data point. This subsection describes the extraction procedure, motivates the correction forms, and explains the uncertainties reported in Table 1.

The imaginary plateau. For each run, we average the imaginary part of the entropy over the four temporal bonds at the centre of the strip. Residual lattice effects produce small oscillations across the interior, particularly at longer times, as shown in Figure 23. Using a single central bond would retain these oscillations, whereas a wider average would include the boundary regions where Eq. (8) no longer applies. Thus, the four-bond average correctly balances these effects. We then take the median over the physical runs at each evolution time to obtain the plateau value per time reported in Table 1.

Extrapolating the plateau. The plateau approaches $\pi c/16$ from above as the finite-time correction decays. We fit $\mathrm{plateau}(T) = a + b\,T^{-1/2}$, where the second term is the Rényi-2 correction predicted for this universality class [23], and obtain the central charge from $c = 16a/\pi$. We test the dependence on this form using a constant fit, a $1/T$ correction, and a two-term refinement of the predicted correction. For the long integrable sequence, the constant fit has the largest residual and gives an asymptote furthest from $\frac{1}{2}$, whereas the predicted form and its refinement give consistent results. The data therefore resolve the finite-time decay. For the shorter interacting windows, all four forms describe the data comparably well but extrapolate to different limits. This inability to distinguish the correction forms limits the interacting estimates, which arises from the accessible time windows alone.

Fitting windows. The upper boundary of each fitting window is the last time for which the sequence remains physical. For the spectral route, it is set by the phase-increment criterion of Section 6.3. For the entropy route, it is set by the ensembles described in Section F.5, which cease to produce physical profiles at a coupling-dependent time. The lower boundary determines when the asymptotic form becomes reliable. The relevant scale is the strip width in conformal units, $v_\infty T$, rather than T alone, and we require $v_\infty T \gtrsim 8$ for every point included in a fit. This condition excludes data only at the integrable point. There, $v_\infty = 2$, so $T = 2$ and $T = 3$ correspond to widths of only 4 and 6 lattice units. At the interacting couplings, the larger velocities place the first available widths between 8 and 11. Consistently, the $p = 0$ plateau continues to rise from $T = 2$ to $T = 4$ and begins the monotonic decay described by the fit only afterwards. We apply this criterion uniformly, without reference to the predicted conformal value.

The branch constant. We fix the constant in Eq. (6) to $B = -\pi$, as required by $\lambda_0 = \log(-\mu_0)$. The integrable sequence is long enough to test this constraint: when B is allowed to vary in a fit that includes the next order in $1/T$, the result is $B/\pi = -1.0000$, with no additional phase constant at the level of 10^{-4}. By contrast, in the shorter two-term fits, a free B absorbs curvature associated with the central-charge term because the constant, $1/T$, and $1/T^2$ contributions cannot be separated over the available window. We therefore unwrap the phase over the complete sequence before selecting the fitting window, which preserves the absolute branch, and keep B fixed.

The real part of the entropy. Both parts of the generalised entropy depend on c, but they do not provide equally reliable estimates. The real part has slope $c/8$ in the chord variable and includes a non-universal offset s_0. For the Rényi-2 entropy in the Ising universality class, it also receives a finite-time correction governed by the energy operator, which decays as $w^{-1/2}$ [23]. Omitting this contribution gives an apparent central charge well above $\frac{1}{2}$, including at the integrable point where the exact value is known; for the profile in Figure 22 it returns 0.60. The complete fitting form is

$$\operatorname{Re} S_2(t) = s_0 + \frac{c}{8} W(t,T) + a\, w(t,T)^{-1/2}, \tag{G.1}$$

Here $w = e^W$ is the chord distance itself, the argument of the logarithm in Eq. (4). This form describes the profiles well, as shown by the dashed curves in Figure 23. However, it does not provide an independent estimate of c within the accessible range. The logarithmic term and the correction have nearly the same shape over this part of the strip, making c and a strongly degenerate in a simultaneous fit. Figure 22 illustrates this degeneracy: the corrected form with $c = \frac{1}{2}$ and an uncorrected logarithmic fit with $c = 0.60$ both describe the same data. The real part therefore serves as a consistency test. For $T \gtrsim 12$, fixing $c = \frac{1}{2}$ and fitting only s_0 and a describes the profiles as well as a fit with c free, and gives a correction amplitude consistent with the imaginary part. We therefore use the imaginary plateau for the central-charge estimate.

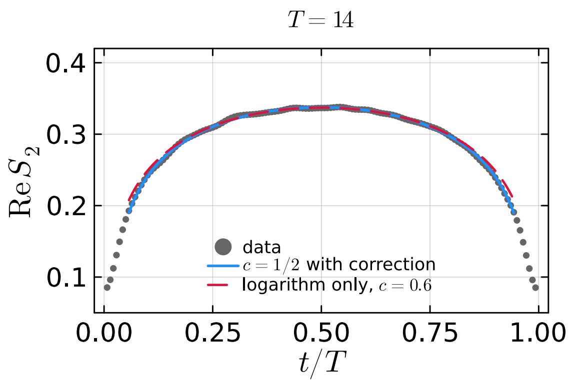


Figure 22: **Central charge from the real part of the entropy.** Integrable-point profile at $T = 14$, shown with the corrected form of Eq. (G.1) for $c = \frac{1}{2}$ and an uncorrected logarithmic fit with c free, which gives $c = 0.60$. Both curves describe the interior data, demonstrating that c and the correction amplitude cannot be separated over this range.

The profiles. Figure 23 presents the generalised temporal entropy at all four couplings. After including the finite-time correction in Eq. (G.1), the real part is consistent with the profile for $c = \frac{1}{2}$. The imaginary part forms an interior plateau above $\pi c/16$. Each row covers the corresponding entropy window in Table 1. These windows decrease rapidly with p, from $T = 22$ at the integrable point to $T = 5$ at $p = 0.5$. The $p = 0.5$ estimate therefore lies furthest above $\frac{1}{2}$, because its sequence ends before the finite-time correction has decayed appreciably.

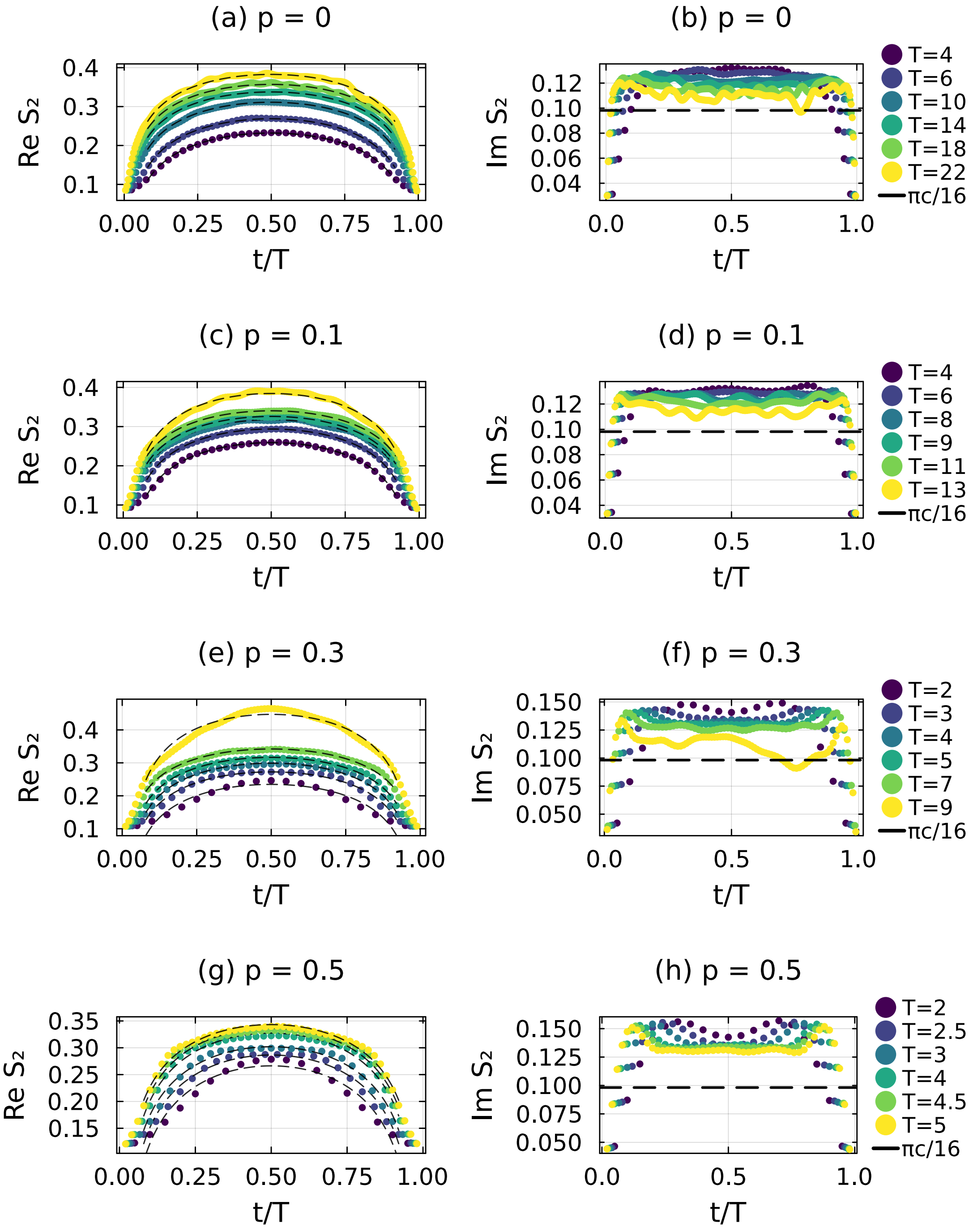


Figure 23: **Generalised temporal entropy at the four couplings.** Each row spans the entropy window for one coupling in Table 1. The left panels show $\mathrm{Re}\, S_2$ as a function of t/T. The dashed curves show Eq. (G.1) for $c = \frac{1}{2}$, with the offset and correction amplitude fitted separately to each profile. The right panels show $\mathrm{Im}\, S_2$; the dashed line marks $\pi c/16 = 0.098$, and the displacement from this value is the finite-time correction.

Near each end of the strip, the imaginary part rises above the interior plateau before decreasing towards the boundary. This behaviour lies outside the regime of Eq. (8), which assumes that both endpoints of the cut are in the bulk. The deviation is confined mainly to the cooling region. Because β_0 remains fixed as T increases, this region occupies a fixed interval and hence a decreasing fraction of the strip. The boundary features therefore remain outside the interior window used to extract the plateau and become relatively less important at longer times.

The phase coefficient and its deficit. At every coupling and for every boundary pair, the $1/T$ coefficient obtained from the spectral fits is systematically 5–10% smaller in magnitude than the asymptotic prediction. Both the central charge and the tower positions are extracted from this coefficient. The deficit therefore accounts for the downward bias in the spectral estimates of Table 1 and the offsets in the fitted positions of Table 11. Two controls identify its sources. First, including the next order in $1/T$ increases the coefficient from 90% to 97% of the predicted value and reduces its dependence on the lower fitting boundary, showing that truncation of the expansion contributes to the deficit. Second, at a fixed fitting window, the coefficient decreases approximately linearly with β_0, from 95% of the prediction at the smallest regulator to 49% at the largest. Extrapolation to $\beta_0 \to 0$ recovers the predicted value (Figure 24), while the production value $\beta_0 = 0.2$ suppresses the coefficient by approximately 12%. The limit $\beta_0 = 0$ cannot be used because the cooling interval produces the regularised conformal boundary state required by the strip construction (Section 2.1). Both effects reduce the fitted coefficient, explaining why the estimates lie systematically below their targets. We retain the same two-term fit at every coupling because the shorter interacting windows cannot support an additional parameter, and treat the residual deficit as a systematic bias rather than applying a pointwise correction.

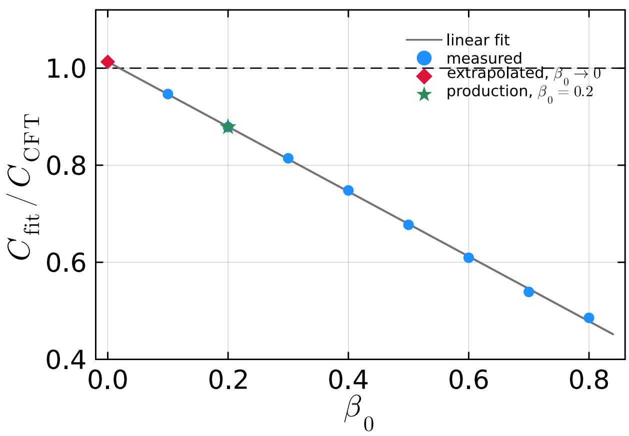


Figure 24: **Fitted phase coefficient as a function of the cooling regulator.** Integrable-point results, expressed as a fraction of the value predicted by Eq. (6). All points use the same fitting window and differ only in N_β. The dependence is approximately linear and extrapolates to within a few per cent of the prediction as $\beta_0 \to 0$. At the production value $\beta_0 = 0.2$, the coefficient lies approximately 12% below the prediction, showing the contribution of the finite regulator to the deficit.

Interpretation of the uncertainties. The entropy estimates in Table 1 include uncertainties because they show measurable variation under resampling. At each time the ensemble provides one plateau value per physical run. We draw a new ensemble of the same size from those values, allowing repeats, take its median, and refit the whole extrapolation; the quoted uncertainty is the standard deviation of the central charge over several hundred such replicas. It is approximately 0.04 at the three interacting couplings and decreases to 0.01 at the integrable point, where the longer sequence prevents any single time point from dominating the fit. The precision is therefore controlled by the number of independent times supporting the extrapolation, rather than by the precision of one plateau value.

The spectral estimates do not have a comparable sampling uncertainty. Independent seeds reproduce the leading eigenvalue to better than 0.1% (Section F.5), and applying the same resampling procedure changes the fitted central charge by less than 10^{-3}, below the final reported digit. Their dominant limitation is instead the systematic bias described above. We therefore report the spectral values without statistical uncertainties and use the two controls to account for their deviation from $\frac{1}{2}$.

G.2 The boundary spectrum and the choice of boundary conditions

The boundary dimensions in Section 7 are extracted using the initial state $|X+\rangle$. More generally, the states at the two ends of the temporal chain define the pair of conformal boundary conditions and thereby determine the operator content allowed on the strip. We test this correspondence by varying the boundary states and comparing the resulting transfer-matrix spectra with the predicted conformal towers, following the same analysis that Bou-Comas et al. carried out at the integrable point [23].

The two boundary conditions determine which operators can appear. If both ends carry the same condition, the identity operator is allowed and the spectrum begins at $x_0 = 0$. Different conditions require a boundary-condition-changing operator, so the lowest level is instead the dimension of the lightest operator connecting them. In the Ising CFT, the relevant primary dimensions are 0 for the identity, $\frac{1}{2}$ for the energy operator, and $\frac{1}{16}$ for the spin operator. Each primary generates descendants at integer level spacings, forming a conformal tower [27, 35].

We represent the three Ising boundary conditions by the product states $|X+\rangle$, $|Up\rangle$, and $|Dn\rangle$. Since $\langle\sigma^z\rangle = 0$, $|X+\rangle$ realises the free condition, whereas $|Up\rangle$ and $|Dn\rangle$ realise the two fixed conditions. Independent choices at the two temporal boundaries give four inequivalent pairs. When the states differ, the network evaluates the transition amplitude $\langle b_2| U(T) |b_1\rangle$ rather than a return amplitude, but the contraction procedure is unchanged.

The phase gaps determine the tower spacings $x_i - x_0$. Table 11 reports the leading spacings at the longest time reached for each pair: $T = 9$ for the free–free and fixed–fixed cases, and $T = 8$ for the more expensive mixed-boundary calculations. Figure 25 shows their evolution with time. All four spectra approach the predicted towers. The two mixed-boundary spectra have the same integer spacings because each contains a single conformal tower generated by its corresponding primary operator.

boundary pair	final T	tower $x_i - x_0$	measured	x_0	measured
free–free	9	$\frac{1}{2},\ \frac{3}{2},\ 2$	0.503, 1.488, 2.016	0	0.002
fixed–fixed	9	2, 3, 4	1.974, 2.982, 3.961	0	0.002
fixed–antifixed	8	1, 2, 3	0.994, 2.001, 3.045	$\frac{1}{2}$	0.474
free–fixed	8	1, 2, 3	0.988, 1.969, 2.980	$\frac{1}{16}$	0.060

Table 11: **Boundary spectra for four pairs of boundary conditions.** Integrable-point results. The spacings are obtained from the phase gaps at the final time listed, while the tower position is obtained by fitting Eq. (6) to the leading phase over the complete sequence. The mixed pairs have identical spacings and are distinguished by x_0.

The spacings alone do not determine the starting dimension x_0. The common non-universal factor $e^{-f\beta - f_s}$ in Eq. (A.13) cancels from μ_i/μ_0, as does the contribution from

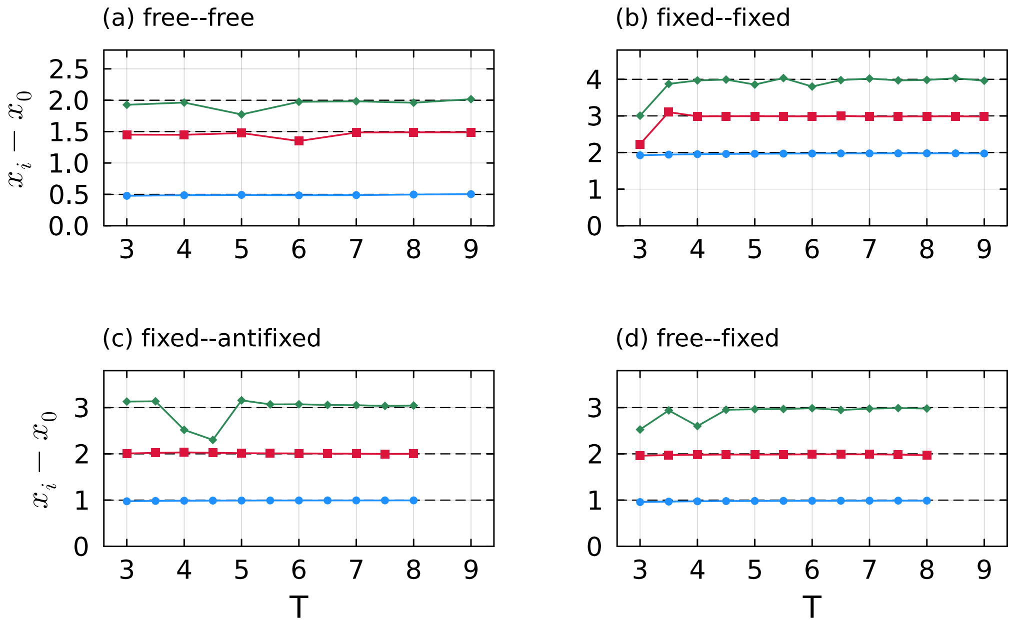


Figure 25: **Boundary towers at the integrable point.** The three leading spacings $x_i - x_0$ are obtained from Eq. (7); dashed lines mark the predicted values. (a) Free–free. (b) Fixed–fixed. (c) Fixed–antifixed. (d) Free–fixed. The two mixed cases have the same spacings and are distinguished by the tower positions in Table 11.

x_0. The tower position is instead encoded in the $1/T$ coefficient of the leading phase. The bulk term $f\beta$ is linear in T, while the real boundary term f_s affects only the modulus (Appendix A.2). The remaining coefficient is

$$\frac{\pi}{v}\left(x_0 - \frac{c}{24}\right). \tag{G.2}$$

At the integrable point, the known values $c = \frac{1}{2}$ and $v = 2$ allow x_0 to be determined by fitting Eq. (6). The two towers predicted to begin at zero give values within 0.002 of zero, while the mixed pairs are assigned to the expected $\frac{1}{2}$ and $\frac{1}{16}$ sectors. The fitted x_0 values therefore distinguish the mixed towers, which have identical spacings.

The small offsets are consistent with the phase-coefficient deficit described in Section G.1. A coefficient that is slightly too small in magnitude displaces every measured position a little, and the displacements seen here are of the size that deficit predicts. They therefore reflect the fitting bias rather than a disagreement with the tower assignments.

The eight-state calculations also resolve degeneracies among conformal descendants. For fixed boundaries, for example, the predicted sequence is $2, 3, 4, 4, 5, 5$, and the numerical spectrum contains pairs near 4 and 5 within the accuracy of the individual estimates. These higher levels support the tower assignments but are not used for precise estimates of individual dimensions.